

AEGIS: A Matrix-Free Operator — Audit, Certify, and Defend Graph Neural Networks

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Abstract

Deploying a graph neural network safely takes two things current tools provide only in isolation: a map of which edge perturbations would break its predictions, and an actionable robustness guarantee. AEGIS delivers both from a single matrix-free object, the *constrained sensitivity matrix* S_c , which records how strongly each edge can move the model’s equilibrium prediction and scales to $N=7,650$ nodes on one GPU. From this one object come three capabilities. AEGIS *audits*: its leading singular direction, column norms, and node margins give an optimal attack, an edge-vulnerability ranking, and per-node safe radii in one pass. It *certifies*: AEGIS-Conformal bounds in closed form the worst-case conformity-score shift, turning split conformal into a distribution-free, sampling-free certificate, sound under exchangeability, that holds at the nominal level under the attack it certifies, and stays non-vacuous on the matched Frobenius ball where smoothing abstains, at $10^4 \times$ lower cost. It *defends*: penalizing $\sigma_1(S_c)$ trades clean accuracy for certified robustness, so the operator that finds the worst attack also tunes the defense (attack and certified radius anticorrelate, -0.65 , 10 seeds). Empirically, this continuous sensitivity predicts single-edge-removal damage across 6 datasets, 7 architectures, and 4 domains (edge-weighted $\tau=0.98$; the sensitivity adds $+0.16$ to $+0.90$ over the weight), and one query inflicts $74\text{--}156 \times$ the per-query equilibrium damage of a 50-step PGD attack.

1 Introduction

Graph neural networks are now deployed where structural errors carry real consequences: fraud accounts evade detection via perturbed transaction edges (Zügner, Akbarnejad, and Günnemann 2018), drug-interaction models misfire under perturbed molecular graphs (Dai et al. 2018), and power grids can miss contingencies when their graph is structurally fragile (Nakiganda et al. 2023) (we audit the model, not the grid’s physics; section 8). A practitioner faces a question current tools leave unanswered: *which edges, if perturbed, would cause predictions to fail, and by how much?*

Each existing thread maps only part of these *adversarial fault lines*: structural attacks rank edges by label-driven gradient search but return no per-node budget or certificate (Zügner, Akbarnejad, and Günnemann 2018; Zügner and Günnemann 2019a; Geisler et al. 2021; Wu et al. 2019); smoothing and certifiers return per-node or collective certificates but neither an edge ranking nor a direction (Bojchevski, Klicpera,

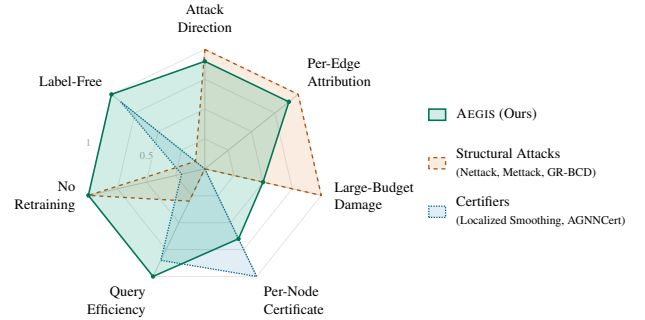


Figure 1: Seven-axis capability radar (frontier semantics; 0 at the centre, 1 at the rim): AEGIS wins no axis outright but is the only method that spans all seven from one object. Defense (the $\sigma_1(S_c)$ penalty) is benchmarked separately (section 5.3); axis definitions and per-method scoring are in appendix F.2.

and Günnemann 2020; Bojchevski and Günnemann 2019; Schuchardt et al. 2023, 2021; Li and Wang 2025; Zügner and Günnemann 2019b); and robust-architecture defenses harden models without surfacing which edges are vulnerable (Zhu et al. 2019; Zhang and Zitnik 2020; Jin et al. 2020). Figure 1 makes this concrete on seven capability axes: attacks and certifiers each cover only their own corner, whereas only AEGIS spans all seven. No prior method audits, certifies, and defends in one matrix-free pass; AEGIS (Adversarial Evaluation of Graph Integrity via Sensitivity) does, from a single object, the constrained sensitivity matrix S_c (section 3). S_c records how strongly each edge can push the equilibrium prediction, so its leading singular direction, column norms, and per-node margins deliver the three diagnostics from one randomized SVD, and the same bound upgrades each per-node radius into a distribution-free certificate (section 4.1). Two scope notes: the closed-form break characterisation (Theorem 1) holds for contractive implicit GNNs, while S_c extends to any GNN with continuous edge-weight message passing (section 5.1); the threat model is edge deletion and weight perturbation (section 2).

Contributions. (1) Constrained sensitivity operator. S_c specialises equilibrium IFT sensitivity (Koh and Liang 2017; Gould, Hartley, and Campbell 2021) to *structural* edge perturbations via P_c ; one matrix-free query reads the SVD-optimal direction, per-edge rankings, and per-node radii together, where no prior object yields all three (matching a dense σ_1

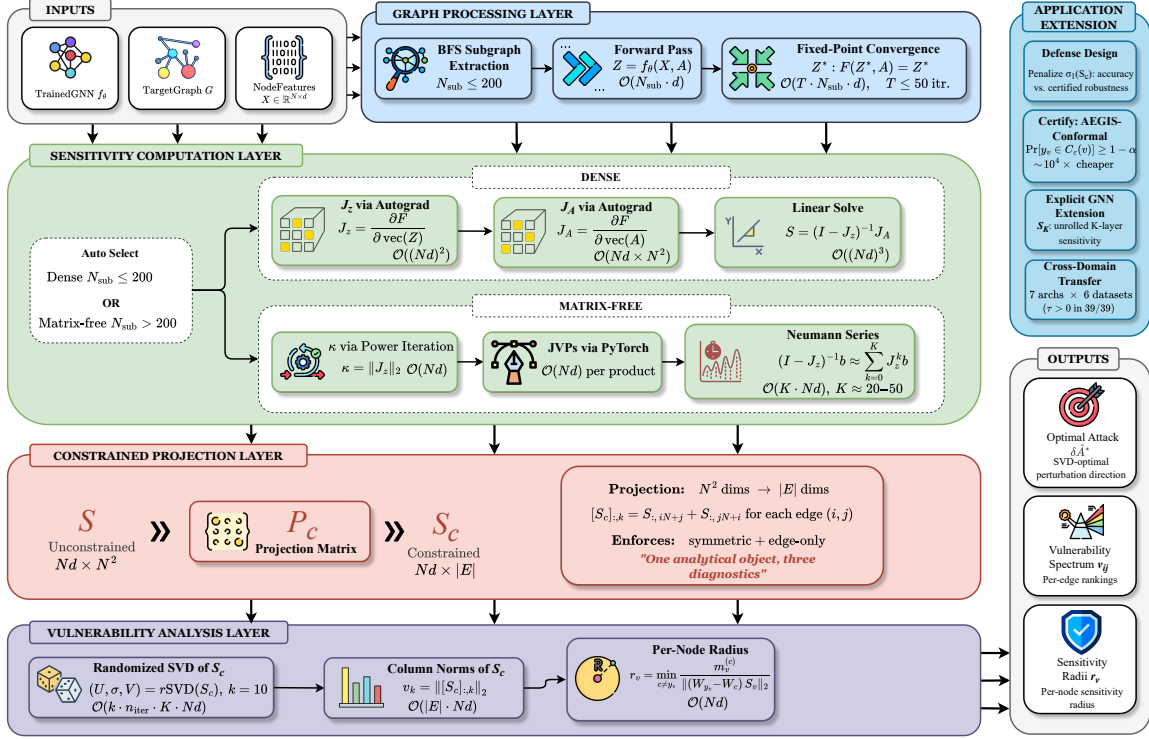


Figure 2: The four-stage AEGIS pipeline (section 3): (1) graph processing (BFS subgraph or full-graph pass); (2) sensitivity (dense Jacobian for $N \leq 200$, else matrix-free Neumann/JVP); (3) constrained projection $N^2 \rightarrow |E|$ via P_c ; (4) analysis (randomised SVD, S_c column norms, per-node radii).

to 0.03% at $N=200$, scaling to $N=7,650$). **(2) Certification from sensitivity.** The same bound yields two guarantees: AEGIS-Conformal, a *distribution-free* certificate (sound under exchangeability) with coverage $\geq 1-\alpha$ over the ε -ball, holding under worst-case attack and non-vacuous where the deterministic radius is thin (section 4.1); and, for contractive models, a constant-factor two-sided characterisation of the breaking point $\varepsilon_{\text{crit}} = (1-\kappa)/\|W\|_2$ in the contraction factor $\kappa = \|J_z\|_2$, where certificate the *measured* break exceeds by $2-9 \times$ (10 seeds). **(3) Coupled defense.** Penalizing $\sigma_1(S_c)$ blunts both AEGIS’s attack and an independent GR-BCD attacker more accuracy-efficiently than generic Lipschitz control, and one knob moves attack, certificate, and that attacker together, so the coupling is operational, not definitional ($-0.65, 10/10$ seeds; section 5.3). **(4) Empirical evaluation** (420 runs): four-quadrant attack comparison, head-to-head vs. GR-BCD/PR-BCD (Geisler et al. 2021) and complementary positioning against AGNNCert (Li and Wang 2025) (appendix F.2), continuous-to-discrete transfer (fig. 6), and a fraud audit (section 6).

2 Preliminaries and Threat Model

Let $G = (V, E, A)$ be an undirected graph with $N = |V|$ nodes, adjacency $A \in \mathbb{R}^{N \times N}$, features $X \in \mathbb{R}^{N \times d_{\text{in}}}$, and symmetric normalized adjacency $\hat{A} = D^{-1/2}(A + I)D^{-1/2}$. Throughout, ϕ is the activation, $\sigma_i(\cdot)$ the i -th singular value, $\text{vec}(\cdot)$ column-major vectorization.

IGNN and IFT. AEGIS analyses models that settle to an equilibrium, because that fixed point is what exposes structural sensitivity in closed form. A DEQ-GNN (Bai, Kolter, and Koltun 2019; Bai, Koltun, and Kolter 2020) iterates a weight-tied F_θ to a fixed point; the IGNN instantiation (Gu et al. 2020) uses

$$F(Z, A) = \phi(\hat{A}ZW^\top + X_{\text{proj}}), \quad (1)$$

with $W \in \mathbb{R}^{d \times d}$ spectral-normalized (Miyato et al. 2018) so $\kappa \leq \|\hat{A}\|_2 \|W\|_2 < 1$, ReLU ϕ , and a learned projection X_{proj} . Training uses implicit differentiation (Gould, Hartley, and Campbell 2021; Fung et al. 2022; Bai, Koltun, and Kolter 2021); a linear head $f(z^*) = W_{\text{cls}} z^* + b$ gives class predictions. At equilibrium, $G(z^*, A) = z^* - F(z^*, A) = 0$ defines z^* as an implicit function of A , with IFT sensitivity

$$\frac{\partial z^*}{\partial p} = (I - J_z)^{-1} \frac{\partial F}{\partial p} \Big|_{z^*}. \quad (2)$$

Equation (2) is the lever AEGIS pulls: the resolvent $(I - J_z)^{-1}$ acts as a gain bounded by $1/(1 - \kappa)$, so as the model nears loss of contractivity ($\kappa \rightarrow 1$) its equilibrium moves ever more violently; section 4 makes this phase transition precise, refined by the pseudospectral index $\eta = \|(I - J_z)^{-1}\|_2 (1 - \rho) \geq 1$ (Trefethen and Embree 2005).

Threat model. A white-box attacker with full access to the trained IGNN perturbs the normalized adjacency

$$\hat{A}' = \hat{A} + \delta \hat{A}, \quad \|\delta \hat{A}\|_F \leq \varepsilon, \quad (3)$$

with $\delta\hat{A}$ symmetric ($\delta\hat{A}=\delta\hat{A}^\top$, preserving the undirected graph), edge-only ($[\delta\hat{A}]_{ij}=0$ for $(i,j) \notin E$, so no new edges), and continuous in $\mathbb{R}$. AEGIS targets edge deletion and weight perturbation; insertion attacks (Nettack-class (Zügner, Akbarnejad, and Günnemann 2018; Zügner and Günnemann 2019a) and node injection (Sun et al. 2020)) would enlarge the basis to $\bar{E} \supseteq E$ but leave the S_c construction intact, and discrete deletions connect to the continuous model through Proposition 3. Unlike input-feature perturbation (El Ghaoui et al. 2021; Revay, Wang, and Manchester 2020), structural perturbation changes the operator itself: z^* depends on A through both $J_z=\text{diag}(\phi')(\hat{A}\otimes W)$ and $J_A=\partial F/\partial \text{vec}(A)$, motivating S_c (section 3).

3 The AEGIS Operator Construction

The previous section isolated the two Jacobians, J_z and J_A , through which the adjacency moves the equilibrium. AEGIS packages them into one operator, the constrained sensitivity matrix S_c , never formed explicitly but only applied to vectors (algorithm 1 and fig. 2).

The constrained sensitivity operator. S_c is the full sensitivity S cut down to the perturbations the threat model permits: one column per existing edge, symmetric under $(i,j) \leftrightarrow (j,i)$. The edge basis has unit Frobenius norm, so $\sigma_1(S_c)$ matches the budget $\|\delta\hat{A}\|_F=\varepsilon$; the $N^2 \rightarrow |E|$ restriction P_c is the standard duplication-matrix reduction (Magnus and Neudecker 2019). One randomized SVD of S_c yields the three audit readings together, the optimal attack direction (Proposition 1), the per-edge ranking $v_{ij}=\|[S_c]_{:,k}\|_2$, and the per-node radius r_v (Proposition 2); the same operator then *certifies* (section 4.1) and *defends* (section 5.3).

Proposition 1 (Maximally Sensitive First-Order Structural Perturbation). *The perturbation δA^* maximizing $\|\Delta z^*\|$ to first order subject to $\|\delta A\|_F \leq \varepsilon$ is $\delta A^* = \varepsilon \cdot \text{reshape}(v_1, N \times N)$, where v_1 is the leading right singular vector of S , with shift $\varepsilon \cdot \sigma_1(S)$. Restricted to the threat model’s feasible set (symmetric, edge-supported; section 2) the maximizer is instead the leading right singular vector of the constrained S_c in the edge basis $\{b_k\}$, with shift $\varepsilon \sigma_1(S_c) \leq \varepsilon \sigma_1(S)$; the unconstrained $\text{reshape}(v_1)$ is generally infeasible.*

This direction maximises hidden-state shift, not label change, so it pinpoints the most dangerous use of the budget, yet still aligns empirically with classification damage (section 5.1).

Matrix-free operator. Scale rests on never forming S_c : every reading is a product $S_c v = (I - J_z)^{-1} J_A P_c v$, applied in $O(Nd)$ memory rather than the $O((Nd)^2)$ of a dense S_c . Contractivity ($\kappa < 1$) expands the resolvent $(I - J_z)^{-1}$ as a geometric Neumann series ($K \in [20, 50]$ for $\kappa < 0.8$), each term a forward-mode JVP that applies P_c implicitly; only for $N \leq 200$ do we form J_z, J_A directly. A randomized SVD (Halko, Martinsson, and Tropp 2011) returns the leading (σ_1, v_1) , symmetrised to keep the attack feasible. One query suffices because the spectrum is well separated: fig. 3 shows $|S_c|$ on a Cora ego-graph with σ_1 a clear 43% above σ_2 (gap 0.39–0.50 across the suite). Removing the $O((Nd)^3)$ dense

Algorithm 1: AEGIS matrix-free vulnerability analysis. S_c is queried but never formed; the dense fallback ($N \leq 200$) builds S_c and runs an exact SVD.

Require: F , fixed point z^* , $\hat{A}$, edge set E , budget ε , rank k , Neumann tolerance tol ; (opt.) head W , margins $m_v^{(c)}$.

Ensure: $\sigma_1(S_c)$, $\delta\hat{A}^*$, $\{v_{ij}\}$, $\{r_v\}$, $\varepsilon_{\text{crit}}$.
1: $\kappa \leftarrow \|J_z\|_2$; $K \leftarrow \lceil \log(1/\text{tol}) / \log(1/\kappa) \rceil$
2: $S_c v := (\sum_{j=0}^K J_z^j) J_A P_c v$ {matrix-free}
3: $(\sigma_1, v_1) \leftarrow \text{rSVD}(S_c, k)$ (Halko, Martinsson, and Tropp 2011)
4: $\delta\hat{A}^* \leftarrow \varepsilon \text{sym}(P_c v_1) / \|\text{sym}(P_c v_1)\|_2$
5: $v_{ij} \leftarrow \|S_c e_{ij}\|_2$ for each $(i, j) \in E$
6: $r_v \leftarrow \min_{c \neq y_v} m_v^{(c)} / \|(W_{y_v} - W_c) S_{c,v}\|_2$ {head only}
7: $\varepsilon_{\text{crit}} \leftarrow (1 - \kappa) / \|W\|_2$ {Theorem 1}
8: **return** $(\sigma_1, \delta\hat{A}^*, \{v_{ij}\}, \{r_v\}, \varepsilon_{\text{crit}})$

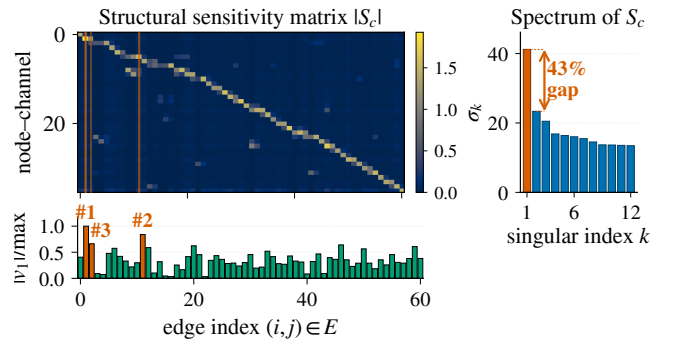


Figure 3: $|S_c|$ on a 50-node Cora ego-graph (IGNN, $N=50$, $d=16$, $|E|=61$, seed 0). Vermilion: top-3 edges by $|v_1|$ (SVD-optimal direction). Sidebar: singular spectrum ($\sigma_1=41.2$, 43% gap to σ_2).

Table 1: One AEGIS query yields all four audit artifacts at the cost of the single cheapest tool and $\sim 700\times$ below the union of separate ones (Cora, 10 seeds, RTX 4090, 50-node subgraph; smoothing $N=10^4$). Artifacts: attack **Direction**, per-edge **Ranking**, per-node **Radius**, **Certificate**.

Method	Time (s)	$\times$ AEGIS	Artifacts			
			Dir	Rank	Rad.	Cert
AEGIS one query	0.24	1	•	•	•	•
GR-BCD (125 ep)	0.19	0.8	•	•	–	–
PR-BCD (125 ep)	2.21	9	•	•	–	–
Smoothing (10^4)	169	695	–	–	–	•
Union	169	696	•	•	–	•

solve lifts the $N \approx 300$ ceiling to $N=7,650$ (appendix F.4). The same query returns all four artifacts in 0.24 s (table 1), the cost of the cheapest single tool and $\sim 700\times$ below their union, which still omits the per-node radius.

4 Adversarial Equilibrium Theory

Section 3 read three diagnostics off S_c ; we now justify them, starting with the safety boundary, a phase transition in ε : the prediction moves boundedly with the graph while ε is small, grows arbitrarily sensitive as ε nears the critical budget $\varepsilon_{\text{crit}}$, and can break past it. Theorem 1 makes this precise and puts

$\varepsilon_{\text{crit}}$ in closed form.

Theorem 1 (Vulnerability Characterization for Contractive Implicit GNNs). *Let $F_\theta(Z, \hat{A}) = \phi(\hat{A}ZW^\top + X_{\text{proj}})$ be an IGNN-class operator satisfying:*

- (A1) ϕ is ReLU or any 1-Lipschitz activation; nonsmoothness at zero is handled via the conservative IFT (Bolte and Pauwels 2021) (proof below).
- (A2) $\|W\|_2 \leq c$ via spectral normalization.
- (A3) $\|J_z\|_2 \leq \kappa < 1$ verified post-training ($\kappa=0.14\text{--}0.59$ across our suite; table 2). (A2) does not imply (A3); we report the trained κ so practitioners can audit, and AEGIS still returns rankings and attacks if (A3) fails, its guarantees then scoped to the contractive regime.

Define two budgets,

$$\varepsilon_{\text{glob}} = \max\left(0, \frac{1}{\|W\|_2} - \|\hat{A}\|_2\right) \leq \varepsilon_{\text{crit}} = \frac{1 - \kappa}{\|W\|_2}. \quad (4)$$

The smaller $\varepsilon_{\text{glob}}$ is conservative but unconditional: a ReLU mask can only shrink a spectral norm, so the perturbed map contracts under every activation pattern once $(\|\hat{A}\|_2 + \varepsilon)\|W\|_2 < 1$, fixing a globally unique equilibrium. The larger $\varepsilon_{\text{crit}}$ reads the same threshold off the trained factor $\kappa = \|J_z\|_2 \leq \|\hat{A}\|_2 \|W\|_2$, the budget at which the first-order rate $\kappa + \varepsilon\|W\|_2$ reaches 1. It is the data-dependent scale used below, equal to $\varepsilon_{\text{glob}}$ only when all units are active. Perturbations $\delta\hat{A}$ with $\|\delta\hat{A}\|_F = \varepsilon$ then pass through three regimes:

(a) **Subcritical** ($\varepsilon < \varepsilon_{\text{crit}}$): the linearization stays contractive and the equilibrium moves only linearly with the budget,

$$\|\Delta z^*\|_F \leq \sigma_1(S)\varepsilon + O(\varepsilon^2), \quad (5)$$

with $S = (I - J_z)^{-1}J_A$ and $\sigma_1(S) \leq \|J_A\|_{\text{op}}/(1 - \kappa)$; below the stricter $\varepsilon_{\text{glob}}$ this fixed point is provably the globally unique one, empirically so up to $\varepsilon_{\text{crit}}$ (Remark 2).

(b) **Critical** ($\varepsilon \rightarrow \varepsilon_{\text{crit}}$): As an eigenvalue of J'_z approaches 1, F stops contracting along that mode and the resolvent gain diverges, $\|(I - J'_z)^{-1}\|_2 \geq 1/\min_i |1 - \lambda_i(J'_z)|$, set by the vanishing spectral gap to 1. The rate is $\Omega(1/(\varepsilon_{\text{crit}} - \varepsilon))$ for normal J'_z ; in general $\varepsilon_{\text{crit}}$ lower-bounds the divergence threshold, the slack being the nonnormality η ($\eta \leq 2.47$ for ReLU; Observation 1 and Remark 1).

(c) **Supercritical** ($\varepsilon \geq \varepsilon_{\text{crit}}$): The contraction certificate no longer applies ($\|J'_z\|_2$ may exceed 1); $\varepsilon < \varepsilon_{\text{crit}}$ is a sufficient boundary that trained IGNNs sit well inside ($2\text{--}4\times$ spectral-radius margin to $\rho=1$, appendix F.4).

Proof in section B.1, a conservative IFT (Bolte and Pauwels 2021) with Neumann resolvent bound $1/(1 - \kappa)$.

The closed-form $\varepsilon_{\text{crit}}$ is only sufficient; the data-dependent S_c tightens it $7\text{--}14\times$, the slack being $\eta \in [1.19, 2.47]$ (Observation 1 and Remark 1).

The same operator bounds the boundary from above too, through W 's nonnormality $g_W = \|W\|_2/\rho(W) \geq 1$.

Theorem 2 (Constant-factor two-sided boundary, all-active). *Under (A1)–(A3) with $\hat{A}$ symmetric and $\varepsilon_{\text{crit}} > 0$, the all-active contraction boundary ε_{br} (the smallest feasible $\|\delta\hat{A}\|_F$ at which $\rho(\hat{A}' \otimes W) \geq 1$) obeys*

$$\varepsilon_{\text{crit}} \leq \varepsilon_{\text{br}} \leq \frac{C}{\beta} \varepsilon_{\text{crit}}, \quad C = g_W \frac{1 + \kappa}{1 - \kappa}, \quad (6)$$

with C dimension-free and computable post-training, and $\beta \in (0, 1]$ the edge-supported Frobenius mass of $\hat{A}$'s Perron mode (positive for any connected graph). The upper side is attained by a rank-one critical-driving attack aligned with that mode; the bracket collapses to equality iff $g_W=1$ and $\beta=1$.

The bracket is loose ($\sim 10\text{--}16\times$) but couples one operator to the certified budget, the attack, and the $\sigma_1(S_c)$ defense; empirically the certificate under-states the measured break by $2\text{--}9\times$ (10 seeds; appendix D).

Proposition 2 (Per-Node First-Order Sensitivity Radius). *For a linear head $f(z) = Wz + b$ and a node v classified as y_v with positive margins $m_v^{(c)} = f_{y_v}(z_v^*) - f_c(z_v^*) > 0$ to every competitor $c \neq y_v$, the first-order sensitivity radius*

$$r_v = \min_{c \neq y_v} \frac{m_v^{(c)}}{\|(W_{y_v} - W_c)S_v\|_2} \quad (7)$$

preserves the classification of v for any $\|\delta\hat{A}\|_F < r_v$, where S_v are the block-rows of S at node v ; substituting $S_{c,v}$ gives the tighter constrained radius $r_v^{(c)} \geq r_v$. Minimizing over all competitors is the distance to the nearest first-order boundary, so the bound holds even when the runner-up changes.

Proof in section B.3. r_v is the smallest margin over two amplifiers, the decision-boundary gap and the equilibrium sensitivity, so high-margin, low-sensitivity nodes earn large radii.

4.1 AEGIS-Conformal: A Distribution-Free Certificate

The radius r_v is a first-order threshold: locally tight, but breachable at larger ε . We upgrade it to a distribution-free guarantee, sound under an exchangeability condition and gated by an explicit coverage test (section E). Split conformal returns a set with $\Pr[y_v \in C(v)] \geq 1 - \alpha$ on clean inputs from a *conformity score*: we use TPS (Sadinle, Lei, and Wasserman 2019) (true-label softmax) and APS (Romano, Sesia, and Candès 2020) (its adaptive variant). Holding this robustly over the ball needs the worst-case score shift, which the same S_c delivers in closed form, a curvature-corrected bound $L_1^c \varepsilon + C_v \varepsilon^2$ replacing the $\sim 10^4$ -sample smoothing the task would otherwise demand. Substituting it into the split conformal of Zargarbashi, Antonelli, and Bojchevski (2023) gives a robust set with

$$\Pr[y_v \in C_\varepsilon(v)] \geq 1 - \alpha \quad \text{for all } \|\delta\hat{A}\|_F \leq \varepsilon. \quad (8)$$

Section 5.2 validates this certificate empirically, checking both nominal-level coverage and the coverage gate under the worst-case attack, and section 5.3 returns to the coupled $\sigma_1(S_c)$ defense.

4.2 Transfer to Discrete Edits and Explicit GNNs

Two bridges carry S_c from the continuous, implicit setting to the discrete, architecture-general one practitioners face; both are proved in section C. The first reads a discrete edge-removal ranking off the continuous score. Deleting an edge is, to first order, a continuous perturbation whose size is the edge’s own weight, so its damage should track the score S_c already assigns it.

Proposition 3 (Continuous-to-discrete transfer). *Under (A1)–(A3), let edge k have weight w_k and score $v_k = \|[S_c]_{:,k}\|_2$, and let d_k be the true damage of removing it. If every single-edge removal is subcritical, then*

$$d_k = w_k v_k + R_k, \quad |R_k| \leq \frac{L_J}{2(1-\kappa)^2} w_k^2, \quad (9)$$

with curvature constant $L_J \leq \|W\|_2^2 \|z^*\|$. The edge-weighted score $w_k v_k$ therefore preserves the brute-force removal order whenever the score gap between two edges beats this quadratic remainder.

The signal is $O(w_k)$ and the error only $O(w_k^2)$, so the ranking is reliable for the small-weight edges of sparse graphs (section 5.1). The second bridge drops the implicit-model assumption: an explicit K -layer GNN has no fixed point, but unrolling its layers yields the same kind of structural Jacobian.

Proposition 4 (Explicit K -layer GNNs). *For $Z_K = g_K \circ \dots \circ g_1(X, A)$ with per-layer Jacobians $J_z^{(k)} = \partial g_k / \partial Z_{k-1}$ and $J_A^{(l)} = \partial g_l / \partial \text{vec}(A)$, the unrolled sensitivity*

$$S_K = \frac{\partial \text{vec}(Z_K)}{\partial \text{vec}(A)} = \sum_{l=1}^K \left(\prod_{k=l+1}^K J_z^{(k)} \right) J_A^{(l)} \quad (10)$$

replaces S throughout: the constrained operator S_c , the per-node radius r_v , and the first-order shift bound $\|\Delta Z_K\|_F \leq \sigma_1(S_K) \|\delta A\|_F$ all carry over unchanged.

Six of our seven architectures thus inherit the full pipeline from S_K ; only the closed-form $\varepsilon_{\text{crit}}$ stays restricted to the contractive implicit subclass.

5 Experiments

Our evaluation follows the three capabilities (sections 5.1 to 5.3), then closes the loop on a deployed fraud detector (section 6).

Setup. We evaluate AEGIS on 6 datasets across 4 domains, 10 seeds each (PyTorch (Paszke et al. 2019); $\pm$ are seed std). The IGNN (Gu et al. 2020) of eq. (1) uses $W \in \mathbb{R}^{64 \times 64}$ spectral-normalised (Miyato et al. 2018) with Adam (lr 0.01). Datasets: Cora, Citeseer, Pubmed (Sen et al. 2008), Amazon Photo (Shchur et al. 2018), WikiCS (Mernyei and Caron 2020), Amazon Fraud (Dou et al. 2020); baselines: smoothing (Gaussian $\sigma=0.01$), Mettack, GR-BCD/PR-BCD (Geisler et al. 2021). Full tables, ablations, timing, and reproducibility are in appendices F and H.

5.1 Audit: One Query Finds the Worst Attack

Two prerequisites must hold: the trained models are contractive, and AEGIS’s perturbations beat chance. Table 2 confirms

Table 2: Per-dataset IGNN characterisation (10 seeds). κ verifies (A3); $\varepsilon_{\text{crit}}$ is the local critical budget; AtkAdv is AEGIS/random damage; Cert is the fraction with $r_v > 0.05$ at $\varepsilon=0.05$.

Dataset	Acc	κ	$\varepsilon_{\text{crit}}$	AtkAdv	Cert
Cora	77.5 $\pm$ 1.7	.33 $\pm$.14	.66 $\pm$.15	3.6 $\pm$.6	.83 $\pm$.10
Citeseer	66.0 $\pm$ 0.7	.59 $\pm$.02	.41 $\pm$.02	4.1 $\pm$.5	.94 $\pm$.05
Pubmed	78.9 $\pm$ 0.7	.41 $\pm$.11	.59 $\pm$.11	3.2 $\pm$.4	.76 $\pm$.18
Amazon	94.8 $\pm$ 0.3	.14 $\pm$.02	.86 $\pm$.02	3.8 $\pm$.2	.86 $\pm$.08
WikiCS	77.9 $\pm$ 0.4	.34 $\pm$.04	.66 $\pm$.04	3.8 $\pm$.4	.78 $\pm$.04

Table 3: One-query audit vs. iterative attackers: equilibrium damage at $\varepsilon=0.10$ (IGNN, 50-node subgraph, 10 seeds). Cls-PGD: classification-loss PGD; Shift-PGD: IFT-gradient PGD (solver validation, not an independent baseline).

Dataset	AEGIS	Cls-PGD	Shift-PGD	Random
Cora	3.70 $\pm$.79	2.51 $\pm$.48	3.05 $\pm$.70	0.97 $\pm$.25
Citeseer	4.63 $\pm$.71	2.97 $\pm$.42	3.35 $\pm$.57	1.01 $\pm$.15
WikiCS	2.10 $\pm$.22	0.67 $\pm$.11	1.93 $\pm$.20	0.53 $\pm$.09

both, verifying (A3) on every benchmark ($\kappa < 1$) and reporting per-dataset $\varepsilon_{\text{crit}}$ with a 3.2–4.1 $\times$ attack advantage over random. Unless stated otherwise, IGNN runs on 50-node BFS ego-subgraphs around the highest-degree node and reports *local* vulnerability; graph-level assessment uses the matrix-free pipeline (appendix F.4).

One query, not a search. AEGIS spends its whole budget in one analytic query: the leading singular direction of S_c is the perturbation that moves the equilibrium most (Proposition 1), with no optimisation loop. Three checks confirm it tracks a real fault line, not the victim’s gradients. *It transfers:* a separately trained surrogate (no shared gradients) recovers 99% of the one-query equilibrium damage ($\cos=0.99$), versus 44 $\pm$ 4% for a 512-query black-box search. *It is efficient:* one query matches or beats 50-step PGD (Madry et al. 2018) on equilibrium damage at $\varepsilon=0.10$ (table 3), a **74–156 $\times$** per-query advantage, though prediction flips stay under 2% at this budget. *It is faithful:* finite differences reproduce the S_c ranking ($\tau=0.999$).

AEGIS also beats degree, edge-betweenness, and spectral rankings at $\varepsilon=0.01$ (+6–148% AtkAdv) and inflicts 3–10 $\times$ more equilibrium damage than Mettack (Zügner and Günnemann 2019a) in the early-warning regime (149/150 wins, $p < 10^{-43}$; fig. 5). First-order shifts match within 1% at small budgets (fig. 8).

Breach rates. Every breached node satisfies $\varepsilon > r_v$ across all datasets and budgets, validating r_v as a first-order screen; flip fractions (fig. 4) stay under 8% except a right-skewed 27.4% on Pubmed at $\varepsilon=0.20$.

Continuous-to-discrete transfer. The bridges of section 4.2 reach beyond implicit models: for seven K -layer architectures (GCN, SAGE, GIN, APPNP, GAT † (Kipf and Welling 2017; Hamilton, Ying, and Leskovec 2017; Xu et al. 2019b; Klicpera, Bojchevski, and Günnemann 2019; Veličković et al. 2018)) we form S_K by a single reverse-mode autodiff query (finite differences for GAT † , whose attention treats $\hat{A}$ as a discrete mask; Proposition 4) and ask whether the continuous ranking predicts brute-force N-1 removal. Ranking by the edge-weighted score $A_{ij}v_{ij}$ (Proposition 3), fig. 6 reports Kendall τ over 6 datasets and 7 architectures (420

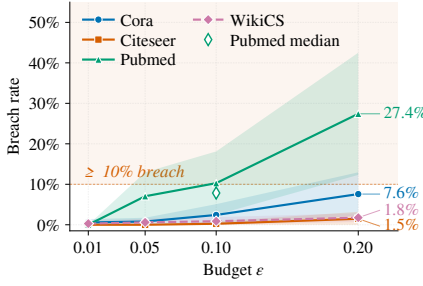


Figure 4: Breach rate (% of nodes flipped under S_c -optimal perturbation) vs. ϵ (10 seeds; mean, 95% CI bands).

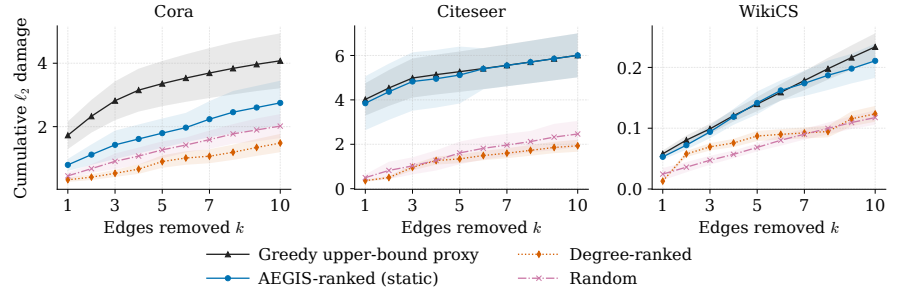
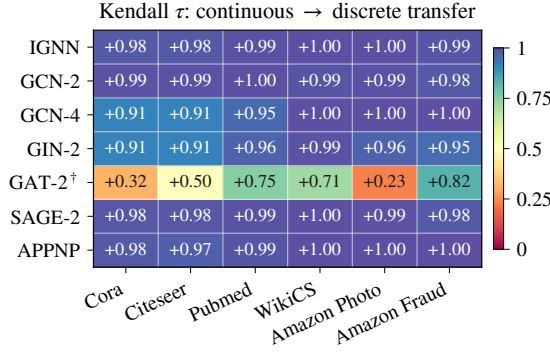


Figure 5: Cumulative damage from sequential discrete edge removal vs. k (IGNN, 50-node subgraph, 10 seeds). AEGIS matches label-aware Greedy on Citeseer/WikiCS and recovers 54–67% of its Cora damage at $k=5-10$ with no label access.



† GAT-2: 2-layer attention; finite-difference sensitivity (attention masks A).

Figure 6: Continuous-to-discrete transfer τ (10-seed means; per-cell sd 0.0003–0.068). GAT-2† transfers better unweighted (appendix F.6).

runs): all 42/42 cells positive, median $\tau=+0.98$ ($p<10^{-5}$). The edge weight carries much of this, since a heavier edge is the larger edit; v_{ij} adds the rest, +0.16 (median) to +0.90 over weight-alone ranking. On full-graph Amazon Photo ($N=7,650$) it reaches $\tau=+0.996$ even at $k\approx 1.00$, past the contractive regime Proposition 3 assumes, so we read it as an empirical regularity, not theory. Per-architecture detail and ablations are in appendices F.4 to F.6.

5.2 Certify: Distribution-Free Coverage

The certificate of section 4.1 holds empirically (table 4; 10 seeds, four datasets). The *gate*, coverage under the worst-case AEGIS attack at magnitude ϵ , sits at the nominal 0.90 at $\epsilon=0.01$ and turns conservative (0.94–1.00) at $\epsilon=0.05$, with informative sets ($\sim 1.0-3.5$ labels) that tighten where the deterministic radius is thin.

Table 5 pits AEGIS-Conformal against randomized smoothing. The decisive gap is cost: the zero-sample S_c bound runs $10^3-10^4\times$ cheaper (appendix F.3). On the matched Frobenius ball it abstains, certifying only on a strictly larger per-coordinate ball.

5.3 Defend: A $\sigma_1(S_c)$ Robustness Knob

The operator that finds the worst attack also tunes the defense: penalizing $\sigma_1(S_c)$ during training drives the dominant sensitivity down, so worst-case attack damage falls while the

Table 4: AEGIS-Conformal coverage (10 seeds; Cora, Citeseer, Pubmed, WikiCS; $\alpha=0.1$, target 0.90). Each ϵ block reports clean coverage / mean set size, then the bold *gate* (coverage under worst-case attack at ϵ). Classes 7/6/3/10; seed std ≤ 0.07 ; sets <1 reflect empty (abstaining) TPS sets.

Dataset	Score	$\epsilon=0.01$		$\epsilon=0.05$	
Cora	APS	0.90 / 1.37	0.90	0.89 / 1.06	0.98
	TPS	0.88 / 0.95	0.90	0.89 / 0.99	0.98
Citeseer	APS	0.92 / 1.50	0.92	0.92 / 1.35	0.97
	TPS	0.91 / 1.20	0.92	0.92 / 1.26	0.96
Pubmed	APS	0.91 / 1.42	0.95	0.92 / 1.52	1.00
	TPS	0.90 / 1.15	0.92	0.90 / 1.15	0.98
WikiCS	APS	0.92 / 3.70	0.94	0.93 / 3.62	0.95
	TPS	0.93 / 3.52	0.94	0.93 / 3.52	0.95

Table 5: AEGIS-Conformal vs. randomized smoothing (Cora, $n=200$, 10 seeds, $\alpha=0.1$, APS). **frob**: matched Frobenius ball ($\sigma=\epsilon/\sqrt{2|E|}$); **per_edge**: larger per-coordinate ball ($\sigma=\epsilon$), favorable to smoothing. Cert: fraction certified; wall-clock at $M=10^4$ samples (AEGIS-Conformal is zero-sample).

Method	Match	ϵ	Cov	Set	Cert	Wall@ 10^4
AEGIS-Conformal	–	0.01	0.90	1.36	–	~ 1 s
RandSmoothing	frob	0.01	1.00	7.00	0.00	7,600 s
RandSmoothing	per_edge	0.01	0.95	1.26	0.96	15,000 s
AEGIS-Conformal	–	0.05	0.90	1.05	–	~ 1 s
RandSmoothing	frob	0.05	1.00	7.00	0.00	10,900 s
RandSmoothing	per_edge	0.05	0.99	2.39	0.77	36,300 s

certified fraction rises (table 6); at $\lambda=3\times 10^{-4}$ this cuts σ_1 by $\sim 10\times$ and lifts the certified fraction to 0.82 ± 0.03 for ~ 4 accuracy points. The two axes answer to different controls: because σ_1 governs the equilibrium response, its penalty blunts both AEGIS’s attack and an *independent* GR-BCD attacker beyond a generic $\|W\|$ cap at matched accuracy (appendix F.5), while the certificate is best served by the cap itself. One knob moves attack, certificate, and that independent attacker together (-0.65 ± 0.12 , 10/10 seeds; fig. 13): the coupling is operational, not definitional.

6 Case Study: Auditing a Fraud Detector

GNN fraud detectors are a prime adversarial target: a flagged account can slip past detection by perturbing a

Table 6: The $\sigma_1(S_c)$ penalty trades clean accuracy for certified robustness (Cora, $c=0.9$ IGNN, 10 seeds). Cert (certified fraction) peaks at $\lambda=3\times 10^{-3}$ before margin collapse.

λ	Acc	$\sigma_1(S_c)$	Cert	Attack Dmg
0	78.1 $\pm$ 1.8	319 $\pm$ 59	.40 $\pm$.08	27.8 $\pm$ 6.3
3×10^{-4}	73.9 $\pm$ 0.8	32.6 $\pm$ 1.7	.82 $\pm$.03	3.1 $\pm$ 0.2
1×10^{-3}	69.0 $\pm$ 0.7	10.7 $\pm$ 0.6	.89 $\pm$.02	1.1 $\pm$ 0.1
3×10^{-3}	61.9 $\pm$ 0.6	3.9 $\pm$ 0.2	.92 $\pm$.02	0.4 $\pm$ 0.0
1×10^{-2}	56.4 $\pm$ 0.6	0.8 $\pm$ 0.1	.86 $\pm$.04	0.1 $\pm$ 0.0

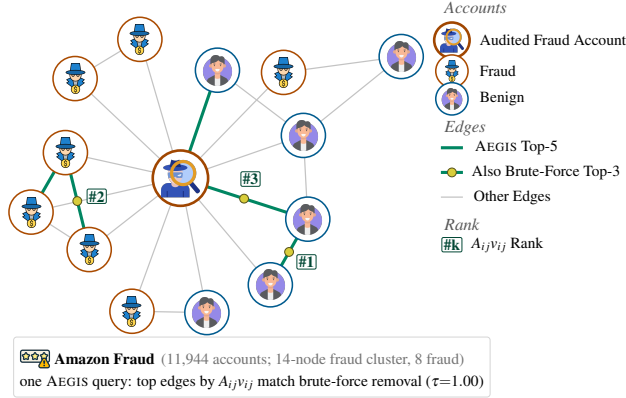


Figure 7: AEGIS auditing an IGNN fraud detector on an Amazon Fraud cluster (Dou et al. 2020). Icons mark fraud / benign accounts; the audited account is the hub. Green edges are the top-5 fault lines by $A_{ij}v_{ij}$; yellow dots mark the three also in the brute-force single-edge-removal top-3.

few behavioural-similarity edges (Zügner, Akbarnejad, and Günnemann 2018; Dou et al. 2020). AEGIS audits one directly, with no labels and no retraining, reading from a *single* S_c query which edges most threaten its predictions. Figure 7 shows the audit on an Amazon Fraud cluster: the edge-weighted ranking $A_{ij}v_{ij}$ (Proposition 3) pinpoints the fault lines, reproducing the brute-force single-edge-removal ranking ($\tau=1.0$ here) at the cost of one query rather than $|E|$ reconvergences. The audit is *non-circular*: both the ranking ($\Delta\tau$ up to +0.90 on the full fraud graph; fig. 6) and the SVD direction (from a separately trained surrogate; section 5.1) transfer, confirming v_{ij} is model-intrinsic, not a gradient artifact. Full walk-through in appendix G.

7 Related Work

Each of three threads supplies at most one of AEGIS’s diagnostics. **Structural attacks** (Nettack, Mettack, GR-BCD/PR-BCD and variants (Zügner, Akbarnejad, and Günnemann 2018; Zügner and Günnemann 2019a; Geisler et al. 2021; Wu et al. 2019; Xu et al. 2019a)) optimise gradients to flip predictions but yield no analytic direction and no per-node radius; sober-look studies (Mujkanovic et al. 2022; Gosch et al. 2023) flag the evaluation pitfalls we adopt (appendix F.5).

Certified robustness and defense. Convex relaxations (Zügner and Günnemann 2019b), smoothing (Bojchevski, Klicpera, and Günnemann 2020; Bojchevski and Günnemann 2019; Schuchardt et al. 2023), collective (Schuchardt et al. 2021) and majority-vote (Li and Wang 2025) certificates, and weight-norm bounds (Abbahaddou

et al. 2024) certify robustness but do not name vulnerability-driving edges; empirical defenses (Zhu et al. 2019; Jin et al. 2020; Zhang and Zitnik 2020) harden models without surfacing per-edge sensitivity. Graph conformal prediction (Zargabashi, Antonelli, and Bojchevski 2023) certifies a *fixed* graph, whereas AEGIS-Conformal covers the adversarial ε -ball via the S_c score-shift bound (section 4.1).

Implicit networks and influence functions. DEQs, IGNN, monotone equilibria, and well-posedness analyses (Bai, Kolter, and Koltun 2019; Gu et al. 2020; Winston and Kolter 2020; El Ghaoui et al. 2021) address *input* sensitivity $\partial z^*/\partial x$; AEGIS targets *structural* sensitivity $\partial Z/\partial A$, applying the Koh–Liang influence lineage (Koh and Liang 2017) (validated against deep-net influence fragility (Basu, Pope, and Feizi 2021) by our finite-difference checks) and implicit differentiation (Lorraine, Vicol, and Duvenaud 2020; Gould, Hartley, and Campbell 2021) to a structural perturbation via P_c rather than to a loss. GNN explainers (Ying et al. 2019) rank edges for *fidelity*, not vulnerability; matrix perturbation theory (Stewart and Sun 1990; Trefethen and Embree 2005) supplies the pseudospectral toolkit; and curvature/oversquashing (Topping et al. 2022; Di Giovanni et al. 2023) and stability bounds (Gama, Bruna, and Ribeiro 2020; Kenlay, Thanou, and Dong 2021) tie edge structure to sensitivity only in aggregate, whereas AEGIS localises it per edge and node.

8 Conclusion

From one matrix-free object, S_c , AEGIS unifies auditing, certification, and defense as readings of a single operator, the same $\sigma_1(S_c)$ that names the worst perturbation also bounding its radius and, penalized, the defense, with a closed-form safety characterisation (Theorem 1) for the contractive implicit case.

Limitations. The radii r_v are first-order; AEGIS-Conformal assumes exchangeability and runs densely at $N=200$; standard GAT, binary-mask architectures, and insertion attacks lie outside scope (sections 2 and 5.1); and the $\varepsilon_{\text{crit}}$ track costs $\sim 6\%$ accuracy. AEGIS audits vulnerability, not physics: a contractive surrogate cannot model voltage collapse, so it complements rather than replaces power-flow contingency screening (Donon et al. 2019; Nakiganda et al. 2023; Varbella, Gjorgiev, and Sansavini 2024).

Disclosure. We propose a 90-day coordinated-notification window: the diagnostic path (r_v, v_{ij}) releases unconditionally as a defender’s tool, while attack-direction reconstruction (algorithm 1) is gated behind ethics review, since it yields the most damaging δA^* (Proposition 1).

References

- Abbahaddou, Y.; Ennadir, S.; Lutzeyer, J. F.; Vazirgiannis, M.; and Boström, H. 2024. Bounding the Expected Robustness of Graph Neural Networks Subject to Node Feature Attacks. In *International Conference on Learning Representations (ICLR)*.
- Bai, S.; Kolter, J. Z.; and Koltun, V. 2019. Deep equilibrium models. In *Advances in Neural Information Processing Systems*, volume 32.
- Bai, S.; Koltun, V.; and Kolter, J. Z. 2020. Multiscale deep

- equilibrium models. In *Advances in Neural Information Processing Systems*, volume 33, 5238–5250.
- Bai, S.; Koltun, V.; and Kolter, J. Z. 2021. Stabilizing equilibrium models by Jacobian regularization. In *International Conference on Machine Learning*, 554–565.
- Basu, S.; Pope, P.; and Feizi, S. 2021. Influence Functions in Deep Learning Are Fragile. In *International Conference on Learning Representations (ICLR)*.
- Bojchevski, A.; and Günnemann, S. 2019. Certifiable robustness to graph perturbations. In *Advances in Neural Information Processing Systems*, volume 32.
- Bojchevski, A.; Klicpera, J.; and Günnemann, S. 2020. Efficient robustness certificates for discrete data: Sparsity-aware randomized smoothing for graphs, images and more. In *International Conference on Machine Learning*, 1003–1013.
- Bolte, J.; and Pauwels, E. 2021. Conservative set valued fields, automatic differentiation, stochastic gradient descent and deep learning. *Mathematical Programming*, 188: 19–51.
- Brody, S.; Alon, U.; and Yahav, E. 2022. How attentive are graph attention networks? In *International Conference on Learning Representations*.
- Dai, H.; Li, H.; Tian, T.; Huang, X.; Wang, L.; Zhu, J.; and Song, L. 2018. Adversarial attack on graph structured data. In *International Conference on Machine Learning*, 1115–1124.
- Di Giovanni, F.; Giusti, L.; Barbero, F.; Luise, G.; Lio, P.; and Bronstein, M. M. 2023. Over-Squashing in Graph Neural Networks: A Comprehensive Survey. In *International Conference on Machine Learning*.
- Donon, B.; Donnot, B.; Guyon, I.; and Marot, A. 2019. Graph neural solver for power systems. In *International Joint Conference on Neural Networks*, 1–8.
- Dou, Y.; Liu, Z.; Sun, L.; Deng, Y.; Peng, H.; and Yu, P. S. 2020. Enhancing Graph Neural Network-Based Fraud Detectors against Camouflaged Fraudsters. In *Proceedings of the 29th ACM International Conference on Information and Knowledge Management (CIKM)*, 315–324.
- El Ghaoui, L.; Gu, F.; Travacca, B.; Askari, A.; and Tsai, A. 2021. Implicit deep learning. *SIAM Journal on Mathematics of Data Science*, 3(3): 930–958.
- Fung, S. W.; Heaton, H.; Li, Q.; McKenzie, D.; Osher, S.; and Yin, W. 2022. JFB: Jacobian-free backpropagation for implicit networks. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, 6648–6656.
- Gama, F.; Bruna, J.; and Ribeiro, A. 2020. Stability Properties of Graph Neural Networks. *IEEE Transactions on Signal Processing*, 68: 5680–5695.
- Geisler, S.; Schmidt, T.; Sirin, H.; Zügner, D.; Bojchevski, A.; and Günnemann, S. 2021. Robustness of graph neural networks at scale. In *Advances in Neural Information Processing Systems*, volume 34, 7637–7649.
- Gosch, L.; Geisler, S.; Sturm, D.; and Günnemann, S. 2023. Adversarial Training for Graph Neural Networks: Pitfalls, Solutions, and New Directions. In *Advances in Neural Information Processing Systems*, volume 36.
- Gould, S.; Hartley, R.; and Campbell, D. 2021. Deep declarative networks. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(8): 3988–4004.
- Gu, F.; Chang, H.; Zhu, W.; Sojoudi, S.; and El Ghaoui, L. 2020. Implicit graph neural networks. In *Advances in Neural Information Processing Systems*, volume 33, 11984–11995.
- Halko, N.; Martinsson, P.-G.; and Tropp, J. A. 2011. Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions. *SIAM Review*, 53(2): 217–288.
- Hamilton, W. L.; Ying, R.; and Leskovec, J. 2017. Inductive Representation Learning on Large Graphs. In *Advances in Neural Information Processing Systems*.
- Jin, W.; Ma, Y.; Liu, X.; Tang, X.; Wang, S.; and Tang, J. 2020. Graph structure learning for robust graph neural networks. In *Proceedings of the 26th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*, 66–74.
- Kenlay, H.; Thanou, D.; and Dong, X. 2021. Stability and Generalisation of Graph Neural Networks via Spectral Analysis. In *International Conference on Machine Learning*, 5412–5422.
- Kipf, T. N.; and Welling, M. 2017. Semi-supervised classification with graph convolutional networks. In *International Conference on Learning Representations*.
- Klicpera, J.; Bojchevski, A.; and Günnemann, S. 2019. Predict then Propagate: Graph Neural Networks meet Personalized PageRank. In *International Conference on Learning Representations*.
- Koh, P. W.; and Liang, P. 2017. Understanding black-box predictions via influence functions. In *International Conference on Machine Learning*, 1885–1894. PMLR.
- Li, J.; and Wang, B. 2025. AGNNCert: Defending Graph Neural Networks against Arbitrary Perturbations with Deterministic Certification. In *USENIX Security Symposium*.
- Lorraine, J.; Vicol, P.; and Duvenaud, D. 2020. Optimizing millions of hyperparameters by implicit differentiation. In *International Conference on Artificial Intelligence and Statistics*, 1540–1552.
- Madry, A.; Makelov, A.; Schmidt, L.; Tsipras, D.; and Vladu, A. 2018. Towards deep learning models resistant to adversarial attacks. In *International Conference on Learning Representations*.
- Magnus, J. R.; and Neudecker, H. 2019. *Matrix Differential Calculus with Applications in Statistics and Econometrics*. Wiley, 3rd edition.
- Mernyei, P.; and Caron, M. 2020. Wiki-CS: A Wikipedia-based benchmark for graph neural networks. In *arXiv preprint arXiv:2007.02901*.
- Miyato, T.; Kataoka, T.; Koyama, M.; and Yoshida, Y. 2018. Spectral normalization for generative adversarial networks. In *International Conference on Learning Representations*.
- Mujkanovic, F.; Geisler, S.; Günnemann, S.; and Bojchevski, A. 2022. Are defenses for graph neural networks robust? In *Advances in Neural Information Processing Systems*, volume 35, 8940–8952.

- Nakiganda, A. M.; Weeraddana, C.; Popli, N.; and van der Hagen, T. 2023. Graph neural networks for fast contingency analysis of power systems. *arXiv preprint arXiv:2310.04213*.
- Paszke, A.; Gross, S.; Massa, F.; Lerner, A.; Bradbury, J.; Chanan, G.; Killeen, T.; Lin, Z.; Gimelshein, N.; Antiga, L.; et al. 2019. PyTorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems*, volume 32.
- Revay, M.; Wang, R.; and Manchester, I. R. 2020. Lipschitz bounded equilibrium networks. In *arXiv preprint arXiv:2010.01732*.
- Romano, Y.; Sesia, M.; and Candès, E. 2020. Classification with Valid and Adaptive Coverage. In *NeurIPS*.
- Sadinle, M.; Lei, J.; and Wasserman, L. 2019. Least Ambiguous Set-Valued Classifiers With Bounded Error Levels. *Journal of the American Statistical Association*, 114(525): 223–234.
- Schuchardt, J.; Bojchevski, A.; Klicpera, J.; and Günnemann, S. 2021. Collective Robustness Certificates: Exploiting Interdependence in Graph Neural Networks. In *International Conference on Learning Representations*.
- Schuchardt, J.; Wollschläger, T.; Bojchevski, A.; and Günnemann, S. 2023. Localized Randomized Smoothing for Collective Robustness Certification. In *International Conference on Learning Representations*.
- Sen, P.; Namata, G.; Bilgic, M.; Getoor, L.; Galligher, B.; and Eliassi-Rad, T. 2008. Collective classification in network data. *AI Magazine*, 29(3): 93–106.
- Shchur, O.; Mumme, M.; Bojchevski, A.; and Günnemann, S. 2018. Pitfalls of graph neural network evaluation. In *Relational Representation Learning Workshop, NeurIPS*.
- Stewart, G. W.; and Sun, J.-g. 1990. *Matrix Perturbation Theory*. Academic Press.
- Sun, Y.; Wang, S.; Tang, X.; Hsieh, T.-Y.; and Honavar, V. 2020. Adversarial Attacks on Graph Neural Networks via Node Injections: A Hierarchical Reinforcement Learning Approach. In *Proceedings of The Web Conference (WWW)*.
- Topping, J.; Di Giovanni, F.; Chamberlain, B. P.; Dong, X.; and Bronstein, M. M. 2022. Understanding over-squashing and bottlenecks on graphs via curvature. In *International Conference on Learning Representations*.
- Trefethen, L. N.; and Embree, M. 2005. *Spectra and Pseudospectra: The Behavior of Nonnormal Matrices and Operators*. Princeton University Press.
- Varbella, A.; Gjorgiev, B.; and Sansavini, G. 2024. Graph Neural Networks for Fast Contingency Analysis of Power Systems. *IEEE Transactions on Power Systems*.
- Veličković, P.; Cucurull, G.; Casanova, A.; Romero, A.; Liò, P.; and Bengio, Y. 2018. Graph Attention Networks. In *International Conference on Learning Representations*.
- Vovk, V.; Gammelman, A.; and Shafer, G. 2005. *Algorithmic Learning in a Random World*. Springer.
- Winston, E.; and Kolter, J. Z. 2020. Monotone operator equilibrium networks. In *Advances in Neural Information Processing Systems*, volume 33, 10718–10728.
- Wu, H.; Wang, C.; Tyshetskiy, Y.; Docherty, A.; Lu, K.; and Zhu, L. 2019. Adversarial examples on graph data: Deep insights into attack and defense. In *International Joint Conference on Artificial Intelligence*, 4816–4823.
- Xu, K.; Chen, H.; Liu, S.; Chen, P.-Y.; Weng, T.-W.; Hong, M.; and Lin, X. 2019a. Topology attack and defense for graph neural networks: An optimization perspective. In *International Joint Conference on Artificial Intelligence*, 3961–3967.
- Xu, K.; Hu, W.; Leskovec, J.; and Jegelka, S. 2019b. How Powerful are Graph Neural Networks? In *International Conference on Learning Representations*.
- Ying, R.; Bourgeois, D.; You, J.; Zitnik, M.; and Leskovec, J. 2019. GNNExplainer: Generating explanations for graph neural networks. In *Advances in Neural Information Processing Systems*, volume 32.
- Zargarbashi, S. H.; Antonelli, S.; and Bojchevski, A. 2023. Conformal Prediction Sets for Graph Neural Networks. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, volume 202 of *Proceedings of Machine Learning Research*, 12292–12318. PMLR.
- Zhang, X.; and Zitnik, M. 2020. GNNGuard: Defending graph neural networks against adversarial attacks. In *Advances in Neural Information Processing Systems*, volume 33, 9263–9275.
- Zhu, D.; Zhang, Z.; Cui, P.; and Zhu, W. 2019. Robust graph convolutional networks against adversarial attacks. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 1399–1407.
- Zügner, D.; Akbarnejad, A.; and Günnemann, S. 2018. Adversarial attacks on neural networks for graph data. In *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2847–2856.
- Zügner, D.; and Günnemann, S. 2019a. Adversarial attacks on graph neural networks via meta learning. In *International Conference on Learning Representations*.
- Zügner, D.; and Günnemann, S. 2019b. Certifiable robustness and robust training for graph convolutional networks. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 246–256.

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A Notation and Preliminaries

This appendix proves every result stated in the main paper and reports the supporting experiments in full. Its purpose is to let a reader reconstruct each claim end to end, so the proofs are written as explicit derivations rather than sketches. To keep them readable we fix all notation once here: every symbol used later appears in table 7, every abbreviation in the glossary below, and the standing assumptions and operator identities that recur across proofs are stated a single time and then invoked by name.

A.1 Symbols

The framework studies one object, the equilibrium of a graph operator, and how it moves when the graph is perturbed. Table 7 lists the symbols in the order a reader meets them: first the operator and its two Jacobians, then the sensitivity matrices read off them, then the two perturbation budgets and the constants that relate them, and finally the classification head and the conformal quantities.

Abbreviations. IFT, implicit function theorem; JVP/VJP, Jacobian–vector and vector–Jacobian products (matrix-free directional derivatives); SVD/rSVD, (randomized) singular value decomposition; PGD, projected gradient descent; BCD, block coordinate descent; MC, Monte Carlo; TPS/APS, threshold and adaptive prediction-set conformity scores. Norms are $\|\cdot\|_2$ (spectral, on operators; Euclidean, on vectors) and $\|\cdot\|_F$ (Frobenius); $\rho(\cdot)$ is the spectral radius and $\otimes$ the Kronecker product.

A.2 Standing assumptions

All proofs operate on the same operator and in the same certified regime, recorded once here.

Definition 1 (Operator and certified regime). *Let $F_\theta(Z, \hat{A}) = \phi(\hat{A}ZW^\top + X_{\text{proj}})$ with equilibrium $z^* = F_\theta(z^*, \hat{A})$. We assume:*

- (A1) Lipschitz activation. ϕ is 1-Lipschitz (ReLU or similar), so $\|\text{diag}(\phi')\|_2 \leq 1$ pointwise; the nonsmoothness of ReLU at 0 is handled by the conservative calculus of Bolte and Pauwels (2021), which supplies an IFT on each linear region.
- (A2) Bounded weight. $\|W\|_2 \leq c$ through spectral normalization. This caps the numerator of g_W but does not by itself bound $\rho(W)$ away from 0 (see appendix D), so (A2) does not imply (A3).
- (A3) Trained contractivity. $\kappa := \|J_z\|_2 < 1$, equivalently $\|\hat{A}\|_2 \|W\|_2 < 1$, verified post-training ($\kappa \in [0.14, 0.59]$ across our suite).
- (A4) All-active operating point. On the operating set every unit is active, $\phi' \equiv 1$, so $J_z = \hat{A} \otimes W$. Only the upper side of the boundary theorem (appendix D) invokes (A4); every certificate uses (A1)–(A3) alone.

Unless stated otherwise we work in the certified regime $\varepsilon < \varepsilon_{\text{crit}}$, where the perturbed operator is still a contraction and the resolvent $(I - J_z)^{-1}$ is well defined.

A.3 The objects the theory reasons about

The operator F_θ maps a node-state matrix Z to a new one by one round of graph propagation and a weight W ; its fixed point z^* is the equilibrium an implicit GNN settles into. Two Jacobians describe how that equilibrium responds to a perturbation. The *state* Jacobian $J_z = \text{diag}(\phi')(\hat{A} \otimes W)$ measures how the next state depends on the current one; under (A3) it is a contraction, so a perturbation injected into the loop decays geometrically. The *structural* Jacobian $J_A = \partial F / \partial \text{vec}(A)$ measures how the operator depends directly on the graph. Differentiating the fixed-point equation $z^* = F_\theta(z^*, \hat{A})$ through the IFT combines the two into the structural sensitivity

$$S = (I - J_z)^{-1} J_A, \quad (11)$$

the single object every diagnostic is read from. The inverse resolvent $(I - J_z)^{-1}$ is what turns a local Jacobian into the equilibrium response: it sums the effect of the perturbation as it circulates around the loop. Restricting the input to feasible (symmetric, edge-supported) perturbations through the projection P_c gives the constrained operator $S_c = (I - J_z)^{-1} J_A P_c$ used in practice.

From S and S_c the paper reads three diagnostics, each proved in a later section: the certified budget $\varepsilon_{\text{crit}}$ below which the prediction cannot break (section B and appendix D); the per-node radius r_v that turns the margin of node v into a safe perturbation distance (section B); and the per-edge score $v_k = \|[S_c]_{:,k}\|_2$ that ranks how much each edge matters (section C). The conformal certificate of section E upgrades the first-order radius into a distribution-free coverage guarantee.

A.4 Matrix-free evaluation of the sensitivity operator

None of the diagnostics forms S explicitly, which is what lets the pipeline scale; the routing is algorithm 1 and we record here why a single matrix–vector rule suffices. Applying S_c to a vector v expands the resolvent as a Neumann series,

$$S_c v = (I - J_z)^{-1} J_A P_c v = \sum_{j=0}^{\infty} J_z^j J_A P_c v, \quad (12)$$

which converges because $\kappa < 1$ (A3). Truncating at depth K leaves a tail bounded by the same contraction factor,

$$\left\| S_c v - \sum_{j=0}^K J_z^j J_A P_c v \right\|_2 \leq \frac{\kappa^{K+1}}{1 - \kappa} \|J_A P_c v\|_2, \quad (13)$$

so the depth needed for a target accuracy grows only logarithmically in $1/\kappa$ ($K \in [20, 50]$ for $\kappa < 0.8$). Each term $J_z^j(\cdot)$ is one forward-mode JVP costing $O(Nd)$ memory rather than the $O((Nd)^2)$ of a dense S , and a randomized SVD returns the leading pair (σ_1, v_1) that the attack and defense use. The well-separated singular spectrum (gap 0.39–0.50) is why one rSVD matches an iterative attacker; the empirical truncation residual $\kappa^{200} \in [10^{-105}, 10^{-48}]$ on the suite confirms (13) is slack at the depths we run.

Table 7: Notation used throughout the appendix. “First used” points to the result that introduces or relies on the symbol.

Symbol	Meaning	First used
<i>Operator and Jacobians</i>		
$\hat{A}$	symmetric normalized adjacency $D^{-1/2}(A+I)D^{-1/2}$	Theorem 1
W	tied weight matrix of the layer	Theorem 1
$\phi, \text{diag}(\phi')$	1-Lipschitz activation; its derivative mask	(A1)
z^*	equilibrium (fixed point) of F_θ	Theorem 1
J_z	state Jacobian $\text{diag}(\phi')(\hat{A} \otimes W)$	Theorem 1
J_A	structural Jacobian $\partial F / \partial \text{vec}(A)$	Theorem 1
κ	contraction factor $\ J_z\ _2 < 1$	(A3)
<i>Sensitivity read off the operator</i>		
S	structural sensitivity $(I - J_z)^{-1} J_A$	Theorem 1
S_c	edge-constrained sensitivity $(I - J_z)^{-1} J_A P_c$	section 3
P_c, P_E	projection onto symmetric, edge-supported perturbations	Theorem 2
$S_v, S_{c,v}$	block rows of S, S_c at node v	Proposition 2
$\sigma_1(\cdot)$	leading singular value	Proposition 1
w_k, v_k	edge weight $[\hat{A}]_{ij}$; edge score $\ [S_c]_{:,k}\ _2$	Proposition 3
d_k	true damage of removing edge k	Proposition 3
<i>Budgets and the constants that relate them</i>		
$\varepsilon_{\text{crit}}$	norm certificate $(1 - \kappa) / \ W\ _2$	Theorem 1
$\varepsilon_{\text{spec}}$	all-active spectral break $1/\rho(W) - \rho(\hat{A})$	Theorem 2
$\varepsilon_{\text{br}}, \varepsilon_{\text{br}}^{\text{all}}$	deployed / all-active contraction-break budget	Theorem 2
$\varepsilon_{\text{reach}}$	measured nonlinear break budget	Remark 2
g_W	nonnormality of W , $\ W\ _2 / \rho(W) \geq 1$	Theorem 2
η	Neumann-to-spectral slack (pseudospectral index)	Observation 1
$\beta = \sigma_E$	edge-supported mass of the Perron mode u_1	Theorem 2
C	bracket constant $g_W(1 + \kappa) / (1 - \kappa)$	Theorem 2
L_J	equilibrium-map curvature, $\leq \ W\ _2^2 \ z^*\ $	Proposition 3
<i>Classification head and conformal prediction</i>		
$f = Wz_v^* + b$	node logits read by the linear head	Proposition 2
$m_v^{(c)}$	logit margin $f_{y_v} - f_c$ to competitor c	Proposition 2
r_v	per-node first-order sensitivity radius	Proposition 2
$\text{score}^{\text{TPS}}, \text{score}^{\text{APS}}$	the two conformity scores	Definition 2
$C(v), \hat{q}$	prediction set; conformal threshold	Definition 2
$L_1^{(c)}, C_v$	first-order and curvature score-shift constants	Lemma 1
α, n_{cal}	target miscoverage; calibration-set size	Theorem 4

B Equilibrium Sensitivity and the Phase Transition

This section proves the three results that turn the operator of section A into a safety boundary: how the equilibrium moves under a perturbation, how far the worst-case slack between the certificate and the true threshold can grow, and the per-node radius that the certificate yields. Each proof restates its claim in brief and then derives it step by step using the notation of table 7.

B.1 Proof of the Phase Transition

Theorem 1 states that, under (A1)–(A3), a structural perturbation of size $\varepsilon = \|\delta \hat{A}\|_F$ passes through three regimes set by the critical budget $\varepsilon_{\text{crit}} = (1 - \kappa) / \|W\|_2$: subcritical ($\varepsilon < \varepsilon_{\text{crit}}$, the equilibrium moves boundedly), critical ($\varepsilon \rightarrow \varepsilon_{\text{crit}}$, its sensitivity diverges), and supercritical ($\varepsilon \geq \varepsilon_{\text{crit}}$, contraction is no longer certified).

Proof. Step 1: the subcritical sensitivity. On each ReLU linear

region F_θ is affine, and the activation pattern at z^* is stable for small $\|\delta \hat{A}\|$ because the set of adjacencies whose equilibrium lands exactly on a region face has Lebesgue measure zero. The conservative calculus of Bolte and Pauwels (2021) therefore supplies an IFT on each region. Differentiating the fixed-point condition $G(z, \hat{A}) = z - F(z, \hat{A}) = 0$ at $(z^*, \hat{A})$ gives the equilibrium response

$$\Delta z^* = (I - J_z)^{-1} J_A \text{vec}(\delta \hat{A}) + O(\|\delta \hat{A}\|^2), \quad (14)$$

which is exactly $S \text{vec}(\delta \hat{A})$ to first order. The resolvent factor is what makes this an equilibrium rather than a one-step response: a perturbation injected into the loop returns attenuated by J_z on each pass, and the total is the geometric sum $\sum_{j \geq 0} J_z^j = (I - J_z)^{-1}$. Since $\kappa = \|J_z\|_2 < 1$ this sum converges with

$$\|(I - J_z)^{-1}\|_2 \leq \sum_{j \geq 0} \kappa^j = \frac{1}{1 - \kappa}, \quad (15)$$

so taking the leading singular value in (14) yields the shift bound $\|\Delta z^*\|_F \leq \sigma_1(S)\varepsilon + O(\varepsilon^2)$ with $\sigma_1(S) \leq \|J_A\|_2/(1-\kappa)$, i.e. (5).

Step 2: two contraction certificates, and why only the smaller one is rigorous. Perturbing the graph moves the state Jacobian to $J'_z = \text{diag}(\phi')(\hat{A}' \otimes W)$, with $\text{diag}(\phi')$ the ReLU mask read at the *perturbed* equilibrium. We ask for which ε the map F is still a contraction, so that the equilibrium is unique and the first-order expansion of Step 1 governs.

The safe answer ignores the mask. A ReLU mask is a 0/1 diagonal projection, so multiplying by it can only shrink a spectral norm, never enlarge it; hence, for *any* activation pattern,

$$\begin{aligned} \|J'_z\|_2 &= \|\text{diag}(\phi')(\hat{A}' \otimes W)\|_2 \leq \|\hat{A}'\|_2 \|W\|_2 \\ &\leq (\|\hat{A}\|_2 + \varepsilon) \|W\|_2, \end{aligned} \quad (16)$$

the last step using $\|\hat{A}'\|_2 \leq \|\hat{A}\|_2 + \|\delta\hat{A}\|_2 \leq \|\hat{A}\|_2 + \varepsilon$. The right-hand side falls below 1 exactly for $\varepsilon < \varepsilon_{\text{glob}} = 1/\|W\|_2 - \|\hat{A}\|_2$. Because the argument never inspected the mask, it survives as the perturbation drives the equilibrium across activation regions; a map of spectral norm below 1 is a Banach contraction, so for every feasible $\varepsilon < \varepsilon_{\text{glob}}$ the perturbed model has a single, *globally unique* equilibrium. This is the rigorous safe radius $\varepsilon_{\text{glob}}$.

The trained models are contractive for a gentler reason than (16) sees. Spectral normalization caps $\|W\|_2 \leq c$, but a normalized adjacency with self-loops has $\|\hat{A}\|_2 = 1$, so $\varepsilon_{\text{glob}} = 1/c - 1$ is small; what holds these models well inside contraction is the ReLU mask, which keeps the trained factor $\kappa = \|J_z\|_2$ far under the worst case $\|\hat{A}\|_2 \|W\|_2$. Reading the threshold off κ gives the larger, data-dependent budget $\varepsilon_{\text{crit}} = (1-\kappa)/\|W\|_2 \geq \varepsilon_{\text{glob}}$: on a *fixed* activation region the perturbed Jacobian is $J'_z = J_z + \text{diag}(\phi')(\delta\hat{A} \otimes W)$, so $\|J'_z\|_2 \leq \kappa + \varepsilon\|W\|_2$, which reaches 1 at $\varepsilon = \varepsilon_{\text{crit}}$.

The phrase “fixed activation region” is where rigor stops. Over a finite budget the equilibrium does not stay on one region; it crosses finitely many ReLU faces before $\varepsilon_{\text{crit}}$, and each crossing can switch on a dormant unit and add a term the single mask $\text{diag}(\phi')$ omits, so $\kappa + \varepsilon\|W\|_2$ is not a valid bound on $\|J'_z\|_2$ across the whole ball; a free choice of mask can push $\|J'_z\|_2$ past 1 well before $\varepsilon_{\text{crit}}$. The two budgets coincide, $\varepsilon_{\text{crit}} = \varepsilon_{\text{glob}}$, only in the all-active case (A4), where $\kappa = \|\hat{A}\|_2 \|W\|_2$ and no region is crossed. For $\varepsilon < \varepsilon_{\text{crit}}$ we therefore claim only what the implicit function theorem gives locally: the equilibrium continues as a unique smooth branch wherever $I - J_z$ is invertible, applied region by region through the conservative calculus of Bolte and Pauwels (2021), together with the first-order bound of Step 1. Whether that branch stays the unique *global* equilibrium all the way to $\varepsilon_{\text{crit}}$ is the empirical regularity of Remark 2 (the measured break lies above $\varepsilon_{\text{crit}}$), whose two scaling gaps remain open; it holds across our suite, but we record it as observation, not

theorem.

Step 3: the critical divergence and why $\varepsilon_{\text{crit}}$ is one-sided. For any matrix M the resolvent $(I - M)^{-1}$ has eigenvalues $(1 - \lambda_i(M))^{-1}$, so

$$\|(I - M)^{-1}\|_2 \geq \rho((I - M)^{-1}) = \frac{1}{\min_i |1 - \lambda_i(M)|}, \quad (17)$$

which blows up when an eigenvalue of M approaches 1. The rate depends on the geometry of J'_z . When J'_z is normal with a dominant real-positive (Perron) eigenvalue equal to $\|J'_z\|_2$, the case for symmetric $\hat{A}$ and W , the spectral gap to 1 is $\min_i |1 - \lambda_i| = 1 - \|J'_z\|_2 = \|W\|_2(\varepsilon_{\text{crit}} - \varepsilon)$ using the within-region linearization $\|J'_z\|_2 = \kappa + \varepsilon\|W\|_2$, giving the $\Omega(1/(\varepsilon_{\text{crit}} - \varepsilon))$ divergence. The certificate is only one-sided because $\|J'_z\|_2 \rightarrow 1$ need not drive the gap to zero: $M = \text{diag}(-s, 0)$ has $\|M\|_2 = s$ yet $\|(I - M)^{-1}\|_2 = 1$, a dominant eigenvalue at $-s$ that stays far from 1. So $\varepsilon_{\text{crit}}$ lower-bounds the true divergence threshold, and the slack is governed by how far J'_z is from normal, the nonnormality η of Observation 1.

Step 4: supercritical. For $\varepsilon \geq \varepsilon_{\text{crit}}$ the bound (16) no longer forces $\|J'_z\|_2 < 1$, the Banach contraction certificate fails, and the iteration may converge to a different fixed point, oscillate, or diverge. $\square$

B.2 A graph-independent bound on the nonnormality

The slack η in Step 3 could in principle grow with the graph; the next observation shows it does not, in the all-active case it is set by W alone.

Observation 1 (Graph-independent nonnormality bound, all-active case). *For $J_z = \text{diag}(\phi')(\hat{A} \otimes W)$ with $\hat{A}$ symmetric and all units active ($\phi' \equiv 1$), the nonnormality obeys $\eta \leq \kappa(V_W)$, where $W = V_W D_W V_W^{-1}$ and $\kappa(V_W) = \|V_W\|_2 \|V_W^{-1}\|_2$ is the eigenvector conditioning of W , independent of $\hat{A}$.*

Proof. Under (A4), $J_z = \hat{A} \otimes W$, whose eigenvalues are the products $\lambda_i(\hat{A})\lambda_j(W)$ and whose eigenvectors are the Kronecker products of those of $\hat{A}$ and W (Stewart and Sun 1990). Symmetric $\hat{A}$ diagonalizes in an orthonormal basis, so its eigenvector matrix has conditioning $\kappa(U_{\hat{A}}) = 1$ and contributes nothing to the joint conditioning; the joint eigenvector matrix therefore has condition number $\kappa(V_W)$. For any diagonalizable $M = V D V^{-1}$ the resolvent obeys $\|(I - M)^{-1}\|_2 \leq \kappa(V)/(1 - \rho(M))$, and the slack relative to the spectral-radius rate $1/(1 - \rho(M))$ is exactly $\kappa(V)$. Applied to J_z this gives $\eta \leq \kappa(V_W)$. $\square$

Remark 1 (Empirical extension to general ReLU patterns). *The bound assumes $\phi' \equiv 1$. For general ReLU patterns the diagonalization argument no longer applies exactly, but the measured nonnormality stays moderate, $\eta \in [1.19, 2.47]$ across the suite (appendix F.5), tracking $\kappa(V_W)$ closely. Graph structure does not amplify it; the modest η traces to the spectral-norm regularization of W , exactly the quantity (A2) controls.*

B.3 Proof of the Per-Node Radius

Proposition 2 gives, for a node v classified as y_v with positive margins $m_v^{(c)} = f_{y_v} - f_c$ to every competitor c , the certified radius $r_v = \min_{c \neq y_v} m_v^{(c)} / \|(W_{y_v} - W_c)S_v\|_2$, below which the prediction cannot change.

Proof. A perturbation of size ε moves the node’s equilibrium by $\Delta z_v^* \approx S_v \text{vec}(\delta \hat{A})$ (Step 1 of section B.1). Because the head is linear, each margin moves by $(W_{y_v} - W_c)\Delta z_v^*$, and Cauchy–Schwarz bounds this by the operator norm of the composed map times the perturbation size,

$$\begin{aligned} |\Delta m_v^{(c)}| &= |(W_{y_v} - W_c)S_v \text{vec}(\delta \hat{A})| \\ &\leq \|(W_{y_v} - W_c)S_v\|_2 \|\delta \hat{A}\|_F. \end{aligned} \quad (18)$$

The prediction of v survives as long as every margin stays positive. The margin to competitor c first reaches zero at $\|\delta \hat{A}\|_F = m_v^{(c)} / \|(W_{y_v} - W_c)S_v\|_2$, and taking the binding (smallest) value over competitors gives

$$r_v = \min_{c \neq y_v} \frac{m_v^{(c)}}{\|(W_{y_v} - W_c)S_v\|_2}. \quad (19)$$

The form is a margin divided by how fast the worst perturbation spends it: the denominator is the product of two amplifiers, the decision-boundary gap $W_{y_v} - W_c$ and the equilibrium sensitivity S_v , so a node earns a large radius exactly when it has a wide margin and a low structural sensitivity. Minimizing over all competitors measures the distance to the nearest first-order decision boundary, so the guarantee holds even if the runner-up class changes along the way; substituting the constrained $S_{c,v}$ for S_v tightens the denominator and yields the larger constrained radius. $\square$

C From Continuous Sensitivity to Discrete Rankings

The sensitivity S_c is a continuous, implicit-model object, while practitioners delete whole edges and use explicit, finite-depth GNNs. Two bridges carry S_c across that gap, and we prove both here: the first shows the continuous score ranks discrete edge removals, the second shows the same construction applies to unrolled explicit networks.

C.1 Continuous-to-Discrete Ranking Transfer

Proposition 3 states that, when every single-edge removal is subcritical, the true removal damage of edge k factors as $d_k = w_k v_k + R_k$ with $|R_k| \leq L_J w_k^2 / (2(1 - \kappa)^2)$, so the edge-weighted score $w_k v_k$ preserves the brute-force removal order whenever the score gap between two edges beats this quadratic remainder. The pairwise condition under which a pair (k_1, k_2) with $v_{k_1} > v_{k_2}$ and $w_{k_1} \geq w_{k_2}$ keeps its order is

$$w_{k_1} < \frac{v_{k_1} - v_{k_2}}{L_J / (1 - \kappa)^2}. \quad (20)$$

Proof. Step 1: a removal is a continuous edge-weight perturbation. Deleting edge k under fixed normalization sets $[\delta \hat{A}]_{ij} = [\delta \hat{A}]_{ji} = -w_k$ with the degree matrix D held fixed, which is exactly the continuous perturbation S_c is built to score. By the first-order sensitivity (14) restricted to the edge, the damage is $\|\Delta z^*\| \approx w_k v_k$ with $v_k = \|[S_c]_{:,k}\|_2$. (Recomputing the normalization instead adds an $O(d_i^{-1})$ rescaling of the incident edges, negligible away from low-degree nodes.) It remains to bound the second-order error R_k .

Step 2: the remainder is governed by one cross term, with nothing omitted. On the interior of a ReLU linear region the activation mask $D = \text{diag}(\phi')$ is constant, so the operator $F(z, \hat{A}) = D(\hat{A} \otimes W)z + Dx$ is bilinear in $(\hat{A}, z)$: linear in z at fixed $\hat{A}$, and linear in $\hat{A}$ at fixed z . Both pure second partials therefore vanish,

$$F_{zz} = \frac{\partial^2 F}{\partial z^2} = 0, \quad F_{AA} = \frac{\partial^2 F}{\partial \text{vec}(A)^2} = 0, \quad (21)$$

and the only surviving second derivative is the adjacency-state cross term F_{zA} . Differentiating the IFT sensitivity $\partial z^* / \partial \text{vec}(A) = (I - J_z)^{-1} J_A$ once more and using (21) to drop every other term gives the full Hessian of the equilibrium map,

$$\frac{\partial^2 z^*}{\partial \text{vec}(A)^2} = (I - J_z)^{-1} (F_{zA} + F_{Az}) (I - J_z)^{-1} J_A. \quad (22)$$

Each summand carries two resolvents $(I - J_z)^{-1}$, a cross partial F_{zA} or F_{Az} proportional to $\|W\|_2$, and J_A proportional to $\|W\|_2 \|z^*\|$ (since $J_A = \partial F / \partial \text{vec}(A) \propto z^* W^\top$). Bounding the resolvents by $1/(1 - \kappa)$ each (eq. (15)) and collecting the weight factors into

$$L_J \leq \|W\|_2^2 \|z^*\|, \quad \|z^*\| \leq \frac{\|X_{\text{proj}}\|}{1 - \kappa}, \quad (23)$$

where the bound on $\|z^*\|$ is the κ -contraction of F at its fixed point (A3), yields $\|\partial^2 z^* / \partial \text{vec}(A)^2\|_2 \leq (1 - \kappa)^{-2} L_J$. The feasible segment $\hat{A} \rightarrow \hat{A} + \delta \hat{A}$ crosses finitely many regions (simultaneous boundary crossings are non-generic, measure zero), and the conservative IFT of Bolte and Pauwels (2021) applies on each, so summing the second-order Taylor remainder across them preserves the bound:

$$|R_k| \leq \frac{1}{2} (1 - \kappa)^{-2} L_J w_k^2. \quad (24)$$

This is the *complete* second-order remainder: the bilinearity (21) leaves no other second-derivative term to bound, so (24) is exact on each region rather than a dominant-term reduction.

Step 3: the score preserves the ranking. With $d_k = w_k v_k + R_k$ from Steps 1–2 and $C := L_J / (2(1 - \kappa)^2)$, two edges differ by

$$d_{k_1} - d_{k_2} \geq (w_{k_1} v_{k_1} - w_{k_2} v_{k_2}) - C(w_{k_1}^2 + w_{k_2}^2). \quad (25)$$

When $v_{k_1} > v_{k_2}$ and $w_{k_1} \geq w_{k_2}$ the first-order gap is positive, and condition (20) is exactly what makes it exceed the summed quadratic remainders, so the sign of $d_{k_1} - d_{k_2}$ matches that of the score gap and the order is preserved. Because the signal scales as $O(w_k)$ and the error only as $O(w_k^2)$,

the ranking is reliable for the small-weight edges of sparse graphs; empirically $|R_k|$ runs $2\text{--}10\times$ below $L_J w_k^2$, and the edge-weighted score reaches Kendall $\tau = +0.996$ against brute-force single-edge removal on full-graph Amazon Photo (section 5.1). $\square$

C.2 Structural Sensitivity for K -Layer Explicit GNNs

Proposition 4 states that an explicit K -layer network $Z_K = g_K \circ \dots \circ g_1(X, A)$, which has no fixed point, still admits the unrolled sensitivity S_K of eq. (10), and that S_K replaces S throughout the pipeline.

Proof. Step 1: unroll and apply the chain rule. Write Z_K as the composition of the K layers and differentiate $\text{vec}(Z_K)$ with respect to $\text{vec}(A)$. Each layer l depends on A both directly, through $J_A^{(l)} = \partial g_l / \partial \text{vec}(A)$, and indirectly through its input Z_{l-1} ; propagating the direct dependence of layer l forward through the later layers' input Jacobians $J_z^{(k)} = \partial g_k / \partial Z_{k-1}$ and summing over l gives

$$S_K = \frac{\partial \text{vec}(Z_K)}{\partial \text{vec}(A)} = \sum_{l=1}^K \left(\prod_{k=l+1}^K J_z^{(k)} \right) J_A^{(l)}, \quad (26)$$

which is (10).

Step 2: the first-order shift bound. Submultiplicativity of the spectral norm applied term by term in (26) gives $\sigma_1(S_K) \leq \sum_{l=1}^K \left(\prod_{k=l+1}^K \|J_z^{(k)}\|_2 \right) \|J_A^{(l)}\|_2$, and a first-order Taylor expansion of Z_K in A turns this into the shift bound $\|\Delta Z_K\|_F \leq \sigma_1(S_K) \|\delta A\|_F + O(\|\delta A\|^2)$.

Step 3: the implicit limit. For a weight-tied model every $\|J_z^{(k)}\|_2 \leq \kappa$, so the sum is geometric,

$$\sigma_1(S_K) \leq \|J_A\|_2 \sum_{j=0}^{K-1} \kappa^j = \|J_A\|_2 \frac{1 - \kappa^K}{1 - \kappa} \xrightarrow{K \rightarrow \infty} \frac{\|J_A\|_2}{1 - \kappa}, \quad (27)$$

recovering the implicit bound of section B.1: unrolling a finite network is the truncated version of the geometric series that the implicit resolvent $(I - J_z)^{-1}$ sums in closed form, so the two agree as depth grows. Because S_K enters the constrained construction S_c and the per-node radius r_v (Proposition 2) in exactly the same place as S , the whole pipeline carries over; only the closed-form $\varepsilon_{\text{crit}}$, which needs the contraction (A3), stays restricted to the implicit subclass. Empirically the explicit models match the implicit tightness (0.99–1.02 across the suite, appendix F.6). $\square$

D The Two-Sided Contraction Boundary

Section B.1 certified safety *below* $\varepsilon_{\text{crit}}$. This section bounds the contraction boundary from *above* as well, proving the full version of Theorem 2: the budget at which the operator first loses contraction sits between $\varepsilon_{\text{crit}}$ and a constant multiple of it. The lower side holds for any 1-Lipschitz ϕ and is the certificate the paper sells; the upper side is proved for the

all-active operator (A4), and the gap between them is a single computable constant. We work under Definition 1, with $\hat{A}$ symmetric and $\varepsilon_{\text{crit}} > 0$.

D.1 Two budgets and the gap between them

Two budgets bound the boundary from opposite sides, the *norm certificate* $\varepsilon_{\text{crit}}$ and the *all-active spectral break* $\varepsilon_{\text{spec}}$:

$$\varepsilon_{\text{crit}} = \frac{1 - \kappa}{\|W\|_2} = \frac{1}{\|W\|_2} - \|\hat{A}\|_2, \quad (28)$$

$$\varepsilon_{\text{spec}} = \frac{1}{\rho(W)} - \rho(\hat{A}). \quad (29)$$

The certificate $\varepsilon_{\text{crit}}$ uses the operator norm $\|W\|_2$, which is the worst case over all directions, so it is conservative; the spectral budget $\varepsilon_{\text{spec}}$ uses the spectral radius $\rho(W)$, which is the actual unstable direction, so it is exact for the all-active operator. The whole gap between them is supplied by the nonnormality of W ,

$$\varepsilon_{\text{spec}} - \varepsilon_{\text{crit}} = \left(\frac{1}{\rho(W)} - \frac{1}{\|W\|_2} \right) + \left(\|\hat{A}\|_2 - \rho(\hat{A}) \right) \geq 0, \quad (30)$$

where the second bracket vanishes because $\hat{A}$ is symmetric, leaving only the term driven by $g_W := \|W\|_2 / \rho(W) \geq 1$.

The quantity g_W is audited per model rather than controlled a priori: spectral normalization (A2) caps $\|W\|_2$ but not $\rho(W)$, so in principle $\rho(W) \rightarrow 0$ at fixed $\|W\|_2$ would send $g_W \rightarrow \infty$. We therefore measure g_W on each trained model ($g_W \in [1.19, 2.47]$ on the suite, with the a-posteriori certificate $g_W \leq \kappa_2(V_W)$, the eigenvector conditioning of W) and report the bracket constant as a function of that value. Every constant below is computable post-training from $\rho(W)$, $\|W\|_2$, $\|\hat{A}\|_2$ and the edge set.

D.2 Edge-feasible critical direction

A break must be driven by a feasible perturbation, one that is symmetric and supported on existing edges. Let P_E be the orthogonal projection onto that subspace $\mathcal{S}_E = \{M = M^\top : M_{ij} = 0 \text{ if } (i, j) \notin E \cup \{(i, i)\}\}$, let u_1 be a unit top eigenvector of $\hat{A}$, and set $g_E := P_E(u_1 u_1^\top)$, $\sigma_E := \|g_E\|_F$, and the unit feasible direction $B := g_E / \sigma_E$. The alignment of this feasible direction with the critical rank-one mode is

$$\beta := \langle u_1, B u_1 \rangle = \frac{\langle u_1, P_E(u_1 u_1^\top) u_1 \rangle}{\sigma_E} = \frac{\|g_E\|_F^2}{\sigma_E} = \sigma_E, \quad (31)$$

using that P_E is an orthogonal projection and $u_1 u_1^\top$ is symmetric. So the alignment β equals the edge-supported Frobenius mass σ_E of the critical direction, which is strictly positive whenever the Perron mode places mass on the edge set, true for every connected graph ($u_1 > 0$ entrywise, $E \neq \emptyset$). The hypothesis $\beta > 0$ is therefore an observable, not an assumption.

D.3 The bracket theorem

Theorem 3 (Constant-factor two-sided bracket for the all-active IGNN contraction boundary, full statement). *Let Definition 1 hold with $\varepsilon_{\text{crit}} > 0$ and $\hat{A}$ symmetric, and let*

$\beta = \sigma_E \in (0, 1]$ be the edge-supported mass of the critical mode. Then:

- (i) **Lower side (any 1-Lipschitz ϕ).** For every feasible $\delta\hat{A}$ with $\|\delta\hat{A}\|_F \leq \varepsilon < \varepsilon_{\text{glob}}$, the perturbed operator is an $\|\cdot\|_2$ -contraction with a unique equilibrium and $\rho(J'_z) < 1$, with no use of (A4); hence $\varepsilon_{\text{br}} \geq \varepsilon_{\text{glob}}$. Under (A4), $\kappa = \|\hat{A}\|_2 \|W\|_2$ gives $\varepsilon_{\text{glob}} = \varepsilon_{\text{crit}}$, so $\varepsilon_{\text{br}} \geq \varepsilon_{\text{crit}}$ and in particular $\varepsilon_{\text{br}}^{\text{all}} \geq \varepsilon_{\text{crit}}$.
- (ii) **Upper side (all-active, requires (A4)).** The feasible rank-one $\delta\hat{A}^* = (\varepsilon_{\text{spec}}/\beta)B \in \mathcal{S}_E$, of size $\|\delta\hat{A}^*\|_F = \varepsilon_{\text{spec}}/\beta$, drives the all-active spectral radius to 1, so $\varepsilon_{\text{br}}^{\text{all}} \leq \varepsilon_{\text{spec}}/\beta$. In the unconstrained-symmetric threat model ($P_E = I$, $\beta = 1$) it is exact: $\varepsilon_{\text{br}}^{\text{all}} = \varepsilon_{\text{spec}}$ with extremizer $\varepsilon_{\text{spec}} u_1 u_1^\top$.
- (iii) **Constant-factor enclosure.** $\varepsilon_{\text{spec}} \leq C \varepsilon_{\text{crit}}$ with the explicit, dimension-free, a-posteriori-computable constant

$$C := g_W \frac{1 + \kappa}{1 - \kappa} = \frac{\|W\|_2}{\rho(W)} \cdot \frac{1 + \|\hat{A}\|_2 \|W\|_2}{1 - \|\hat{A}\|_2 \|W\|_2},$$

so the all-active boundary obeys

$$\boxed{\varepsilon_{\text{crit}} \leq \varepsilon_{\text{br}}^{\text{all}} \leq \frac{C}{\beta} \varepsilon_{\text{crit}}}. \quad (32)$$

The bracket is non-vacuous whenever $\beta > 0$; on the suite $\beta \approx 0.62$, giving $C/\beta \lesssim 16$. The β -free form $\varepsilon_{\text{crit}} \leq \varepsilon_{\text{br}}^{\text{all}} \leq C \varepsilon_{\text{crit}}$ holds exactly for the unconstrained-symmetric model ($\beta = 1$).

- (iv) **Equality conditions.** (a) the endpoints coincide, $\varepsilon_{\text{crit}} = \varepsilon_{\text{spec}}$, iff $g_W = 1$ (W normal), independent of κ, β ; (b) all three coincide, $\varepsilon_{\text{crit}} = \varepsilon_{\text{br}}^{\text{all}} = \varepsilon_{\text{spec}}$, iff $g_W = 1$ and $\beta = 1$; (c) the slack constant is one, $C/\beta = 1$, iff additionally $\kappa \rightarrow 0^+$.

Proof. Setup. Feasible perturbations lie in $\mathcal{S}_E$ with $\|\delta\hat{A}\|_F = \varepsilon$. We use Weyl's inequality, $\rho(\hat{A}') \leq \rho(\hat{A}) + \|\delta\hat{A}\|_2$ and $\|\hat{A}'\|_2 \leq \|\hat{A}\|_2 + \|\delta\hat{A}\|_2$, together with $\|\delta\hat{A}\|_2 \leq \|\delta\hat{A}\|_F = \varepsilon$. Since $\hat{A}$ and the extremal $\delta\hat{A}$ are symmetric, $\rho(\hat{A}') = \|\hat{A}'\|_2$ and Weyl holds with equality along an aligned rank-one mode.

(i) *Lower side.* By (A1) the activation mask is a 0/1 projection, so exactly as in (16) $\|J'_z\|_2 \leq (\|\hat{A}\|_2 + \varepsilon) \|W\|_2$ for every mask, which is below 1 for $\varepsilon < \varepsilon_{\text{glob}}$. A spectral-norm contraction has a unique fixed point and $\rho(J'_z) \leq \|J'_z\|_2 < 1$, so no feasible $\varepsilon < \varepsilon_{\text{glob}}$ breaks contraction: $\varepsilon_{\text{br}} \geq \varepsilon_{\text{glob}}$, using (A1)–(A3) only. Under the all-active operating point (A4) assumed for the upper side, $\kappa = \|\hat{A}\|_2 \|W\|_2$, so $\varepsilon_{\text{glob}} = \varepsilon_{\text{crit}}$ and $\varepsilon_{\text{br}}^{\text{all}} \geq \varepsilon_{\text{crit}}$. The additive gap (30) then gives $\varepsilon_{\text{crit}} \leq \varepsilon_{\text{spec}}$.

(ii) *Upper side, all-active.* Let u_1 be a unit top eigenvector of symmetric $\hat{A}$. Along the rank-one symmetric direction

$t u_1 u_1^\top$ ($\|u_1 u_1^\top\|_F = 1$), $\rho(\hat{A} + t u_1 u_1^\top) = \rho(\hat{A}) + t$ exactly, so under (A4), $\rho(J'_z) = (\rho(\hat{A}) + t)\rho(W)$ reaches 1 at $t = \varepsilon_{\text{spec}}$, giving the unconstrained break budget $\varepsilon_{\text{spec}}$ with extremizer $\varepsilon_{\text{spec}} u_1 u_1^\top$. For edge feasibility, project to $\mathcal{S}_E$ via $B = P_E(u_1 u_1^\top)/\sigma_E$. The map $t \mapsto \lambda_{\max}(\hat{A} + tB) = \max_{\|v\|=1} v^\top (\hat{A} + tB)v$ is a pointwise maximum of affine functions, hence convex, so it lies above its tangent at $t = 0$, $\rho(\hat{A} + tB) \geq \rho(\hat{A}) + \beta t$ for all $t \geq 0$ (this lower direction is what convexity buys; a concave version fails on every random instance). Choosing $\delta\hat{A}^* = (\varepsilon_{\text{spec}}/\beta)B$ drives $\rho(\hat{A}') \geq \rho(\hat{A}) + \varepsilon_{\text{spec}} = 1/\rho(W)$, so $\rho(J'_z) \geq 1$ and $\varepsilon_{\text{br}}^{\text{all}} \leq \|\delta\hat{A}^*\|_F = \varepsilon_{\text{spec}}/\beta$, exact when $\beta = 1$.

(iii) *Closing the bracket.* Dropping $-\rho(\hat{A}) \leq 0$ and using $\rho(\hat{A}) = \|\hat{A}\|_2$, $\varepsilon_{\text{spec}} \leq 1/\rho(W) = g_W/\|W\|_2 = g_W(\varepsilon_{\text{crit}} + \|\hat{A}\|_2)$. Since $\|\hat{A}\|_2 = \kappa/\|W\|_2 = \kappa(\varepsilon_{\text{crit}} + \|\hat{A}\|_2)$ gives $\|\hat{A}\|_2 = \frac{\kappa}{1-\kappa}\varepsilon_{\text{crit}}$, we obtain $\varepsilon_{\text{spec}} \leq g_W \frac{1}{1-\kappa}\varepsilon_{\text{crit}} \leq g_W \frac{1+\kappa}{1-\kappa}\varepsilon_{\text{crit}} = C \varepsilon_{\text{crit}}$. With (i) and (ii) the boxed bracket (32) follows.

(iv) *Equality conditions.* (a) The gap $\frac{1}{\rho(W)} - \frac{1}{\|W\|_2} \geq 0$ is zero iff $g_W = 1$, with κ, β absent. (b) Adding edge-feasibility of the exact extremizer $\varepsilon_{\text{spec}} u_1 u_1^\top$ requires $\beta = 1$, giving $\varepsilon_{\text{crit}} = \varepsilon_{\text{br}}^{\text{all}} = \varepsilon_{\text{spec}}$. (c) Forcing $\frac{1+\kappa}{1-\kappa} = 1$ in addition requires $\kappa \rightarrow 0^+$. $\square$

This recovers and sharpens Theorem 1: the old result is the lower side (i), part (ii) supplies the matching all-active upper side, and (iv) separates the three equality conditions. The characterization is constant-factor (dimension-free), rate-sharp in the normal, edge-aligned corner (iv)(b), with enclosure constant C/β reaching 10–16 $\times$ on the suite (random-instance $\varepsilon_{\text{spec}}/\varepsilon_{\text{crit}} \in [1.02, 18.2]$, median 2.25). Two identities, both numerically exact over thousands of random certified-regime instances, underlie it: the all-active spectral law $\rho(J'_z) = \rho(\hat{A}')\rho(W)$ (A4 only), and the additive gap (30), whose slack is supplied entirely by W 's nonnormality when $\hat{A}$ is symmetric.

D.4 First-order tightness across the budget

The bracket rests on the first-order term being an accurate proxy for the true shift at the budgets we use; fig. 8 confirms this directly. The ratio of the true equilibrium shift to its first-order prediction stays within the operational budget and grows only as ε leaves the linear regime, which is exactly why the realistic-budget analysis below relies on the first-order $\sigma_1(S_c)$ rather than on proximity to criticality.

D.5 Nonlinear break budget: an empirical extension

Remark 2 (Observation O1: nonlinear break budget and the active fraction; empirical, not part of Theorem 2). *The deployed model is ReLU, not all-active, and its measured nonlinear break budget $\varepsilon_{\text{reach}}$ exceeds the all-active $\varepsilon_{\text{spec}}$. Over the 10 preferred seeds (50-node ego subgraph, $\|\hat{A}\|_F = 1.77$):*

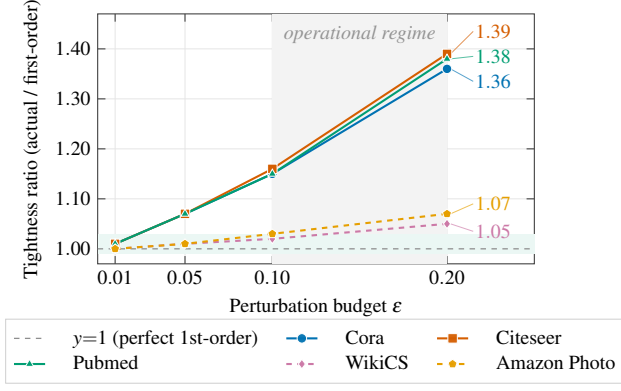


Figure 8: First-order tightness of the equilibrium-shift prediction versus the perturbation budget ε (full graph, mean over 10 seeds). The ratio is the true shift divided by the first-order prediction $\sigma_1(S)\varepsilon$; it is 1 when the linearization is exact and rises as ε grows. Within the operational regime ($\varepsilon \leq 0.10$) every curve stays below 1.16; only at $\varepsilon = 0.20$ do citation graphs (solid) reach 1.36–1.39, while product graphs (dashed) stay within 1.07. This shift-prediction ratio is distinct from the ranking tightness quoted elsewhere (1.00–1.02, appendices F.5 and F.6), which is measured at the operational sub-graph scale; the figure traces the budget dependence that motivates capping ε in that regime.

at $\kappa_0 = 0.5$, $\varepsilon_{\text{reach}}/\varepsilon_{\text{spec}} = 1.41 \pm 0.12$ and $\varepsilon_{\text{reach}}/\varepsilon_{\text{crit}} = 2.17 \pm 0.18$; at $\kappa_0 = 0.9$, $\varepsilon_{\text{reach}}/\varepsilon_{\text{spec}} = 1.51 \pm 0.18$ and $\varepsilon_{\text{reach}}/\varepsilon_{\text{crit}} = 8.72 \pm 1.03$. Empirically $\varepsilon_{\text{reach}} \approx \varepsilon_{\text{spec}}/a$ with a the ReLU active fraction: $a \approx 0.6$ gives $1/a \approx 1.5$ – 1.7 , matching the measured $\varepsilon_{\text{reach}}/\varepsilon_{\text{spec}}$, and composing with the bracket gives the observed envelope $\varepsilon_{\text{reach}}/\varepsilon_{\text{crit}} \lesssim C/a$, spanning roughly [2, 10].

This is a conjecture; two independent proof gaps remain open. (1) Masked-operator spectral scaling (unproved): the relation $\varepsilon_{\text{reach}} \approx \varepsilon_{\text{spec}}/a$ assumes the dominant eigenvalue of the masked Jacobian $D_a(\hat{A}' \otimes W)$ scales as $\approx a \rho(\hat{A}')\rho(W)$, but no such lemma holds in general because the active set correlates with the equilibrium and with u_1 ; this is a mean-field heuristic calibrated to two operating points. (2) Linear-to-nonlinear bifurcation (empirical only): the step from “a linearized eigenvalue reaches 1” to “the true nonlinear fixed point destabilizes” is supported here only empirically across the 10 seeds (a clean simple-pole divergence, critical exponent $\gamma \approx 1.02$ – 1.06). We present $\varepsilon_{\text{reach}} \approx \varepsilon_{\text{spec}}/a$ as an empirical regularity organized by the same two knobs (g_W and a), not a proven extension. The practical consequence is benign: $\varepsilon_{\text{reach}}$ runs 5–11 $\times$ the largest realistic budget ($\varepsilon \leq 0.2$), so realistic-budget robustness is governed by the first-order $\sigma_1(S_c)$ of fig. 8, not by proximity to criticality.

E The AEGIS-Conformal Coverage Bound

This section derives the certificate behind eq. (8): a worst-case conformity-score-shift bound $L_1^c \varepsilon + C_v \varepsilon^2$ and the robust-coverage guarantee it implies. The argument has three parts, each building on the sensitivity already established. Definition 2 fixes the two conformity scores and the two constants. Lemma 1 bounds the worst-case score shift using the first-order margin sensitivity (the Cauchy–Schwarz step of Proposition 2) and the second-order curvature (the complete remain-

Table 8: Reconciliation of the robustness constants. The three “ $\times$ ” ratios that recur in the text are distinct quantities, not estimates of one number: the operating margin is how far the trained model sits inside $\rho = 1$, the empirical conservatism is how far $\varepsilon_{\text{crit}}$ under-states the true nonlinear break, and the proven bracket is the worst-case enclosure constant.

Quantity	Meaning	Source
$\kappa = \ J_z\ _2$	trained contractivity (< 1)	table 2
$\varepsilon_{\text{crit}} = \frac{1-\kappa}{\ W\ _2}$	norm certificate (safe radius)	Theorem 1
$\varepsilon_{\text{spec}}$	all-active spectral break budget	Theorem 2
2–4 $\times$	operating ρ -margin to $\rho = 1$	appendix F.4
2–9 $\times$	empirical $\varepsilon_{\text{reach}}/\varepsilon_{\text{crit}}$ (10 seeds)	Remark 2
10–16 $\times$	proven bracket constant C/β	Theorem 2
$\gamma \approx 1.02$	resolvent critical exponent	Remark 2
$g_W \in [1.19, 2.47]$	nonnormality $\ W\ _2/\rho(W)$	Theorem 2

der of section C.1). Theorem 4 inflates the split-conformal threshold by that shift and reduces to the binary certificate of Zargarbashi, Antonelli, and Bojchevski (2023). We use (A1)–(A3) and operate in the certified regime $\varepsilon < \varepsilon_{\text{crit}}$.

Conformal setup. Fix a target node v with true label $y_v \in [C]$, and let $f = Wz_v^* + b \in \mathbb{R}^C$ be its pre-softmax logits (rows $W_1, \dots, W_C$) with class probabilities $\pi = \text{softmax}(f)$. For a feasible perturbation $\delta \hat{A}$ of size at most ε , write the perturbed logits $f' = Wz_v^{*'} + b$ and probabilities $\pi' = \text{softmax}(f')$, and let $S_{c,v}$ be the block rows of S_c at node v . By the first-order sensitivity (14),

$$\Delta z_v^* = S_{c,v} \delta + R_v, \quad \|\delta\|_2 = \|\delta \hat{A}\|_F \leq \varepsilon, \quad (33)$$

where δ are the edge-basis coordinates of $\delta \hat{A}$ and R_v is the remainder bounded in section C.1. The edge basis is orthonormal, so the coordinate map is an isometry and the Frobenius budget passes through unchanged, which is the normalization that makes $\sigma_1(S_c)$ consistent with $\|\delta \hat{A}\|_F$.

Definition 2 (Conformity scores and the two AEGIS constants). For node v and candidate label $r \in [C]$, define the two conformity scores (higher means more conforming; the prediction set is $C(v) = \{r : \text{score}_r(v) \geq \hat{q}\}$):

$$\text{TPS: } \text{score}_r^{\text{TPS}}(v) = \pi_r, \quad (34)$$

$$\text{APS: } \text{score}_r^{\text{APS}}(v) = 1 - (\rho_r + u_v \pi_r), \quad (35)$$

$$\rho_r := \sum_{c: \pi_c > \pi_r} \pi_c, \quad (36)$$

with $u_v \sim \text{Unif}[0, 1]$ a per-node tie-break drawn once and shared between calibration and test, so it cancels under exchangeability. For each competitor $c \neq r$ let $g_c := f_r - f_c$ be the logit margin. Define the first-order and curvature constants

$$L_1^{(c)} := \|(W_r - W_c) S_{c,v}\|_2, \quad (37)$$

$$C_v := \|W_r - W_c\|_2 \cdot \frac{L_{J,v}}{2(1-\kappa)^2}, \quad (38)$$

where $L_{J,v} \leq \|W\|_2^2 \|z^*\|$ is the per-node curvature of section C.1 and $\kappa = \|J_z\|_2 < 1$ (A3). Equation (37) is the

first-order margin sensitivity; (38) is the curvature constant $L_J/(2(1-\kappa)^2)$ of Proposition 3 scaled by the readout gap. The aggregate node constant in eq. (8) takes the competitor-worst values, $L_1^c = \max_{c \neq r} L_1^{(c)}$ and C_v with the worst $\|W_r - W_c\|_2$.

Remark 3 (Why the gradient is the readout gap, not $\nabla_{z\text{score}}$). The constant (37) uses the logit-margin rows $W_r - W_c$ rather than a softmax-Jacobian row $\nabla_{z\text{score}_r}$, for two reasons. First, both scores depend on the logits only through the margins $\{g_c\}_{c \neq r}$ (softmax sees logits only through differences), so controlling every margin controls π_r and ρ_r and hence both scores (Lemma 1). Second, the margin form needs no bound on softmax curvature: it yields a sound, one-sided score drop without the $1/4$ softmax-Lipschitz slack. The $\nabla_{z\text{score}}$ form is literally correct only for TPS, whereas the margin form covers both.

Lemma 1 (Worst-case conformity-score shift). Assume (A1)–(A3), $\varepsilon < \varepsilon_{\text{crit}}$, and a linear head $f = Wz_v^* + b$. Then for every feasible $\delta\hat{A}$ with $\|\delta\hat{A}\|_F \leq \varepsilon$ and every label r ,

$$|g_c(\hat{A} + \delta\hat{A}) - g_c(\hat{A})| \leq L_1^{(c)}\varepsilon + C_v\varepsilon^2 \quad (39)$$

for every competitor $c \neq r$, and consequently the worst-case drop of the conformity score of label r obeys

$$\begin{aligned} \text{score}_r(v; \hat{A}) - \text{score}_r(v; \hat{A} + \delta\hat{A}) &\leq \Delta_r(\varepsilon), \\ \Delta_r(\varepsilon) &:= \Psi_r(\{L_1^{(c)}\varepsilon + C_v\varepsilon^2\}_{c \neq r}), \end{aligned} \quad (40)$$

where Ψ_r is the monotone map from margin decrements to the induced score decrement. Taking the competitor-worst constants gives the headline $\Delta_r(\varepsilon) \leq L_1^c\varepsilon + C_v\varepsilon^2$.

Proof. Step 1: the linear term (Cauchy–Schwarz). Because the head is affine, $g_c = (W_r - W_c)z_v^* + (b_r - b_c)$, so $g'_c - g_c = (W_r - W_c)\Delta z_v^*$. Inserting (33) splits this into a first-order and a remainder part,

$$g'_c - g_c = (W_r - W_c)S_{c,v}\delta + (W_r - W_c)R_v. \quad (41)$$

The first summand is a fixed covector paired with δ , so Cauchy–Schwarz and $\|\delta\|_2 \leq \varepsilon$ bound it by $L_1^{(c)}\varepsilon$ through definition (37). This is the same step that proves the radius r_v , now read with the readout gap $W_r - W_c$ in place of the full S_v image, so the constant is the rate at which the most exposed margin moves per unit perturbation.

Step 2: the quadratic term (curvature). For the remainder, $|(W_r - W_c)R_v| \leq \|W_r - W_c\|_2 \|R_v\|_2$. By section C.1 the equilibrium-map Hessian is the complete second derivative (22), bounded in operator norm by $(1 - \kappa)^{-2}L_{J,v}$ with $L_{J,v} \leq \|W\|_2^2 \|z^*\|$ finite under the contraction (A3). A second-order Taylor expansion along the feasible segment (finitely many ReLU regions, conservative IFT on each) therefore gives $\|R_v\|_2 \leq \frac{1}{2}(1 - \kappa)^{-2}L_{J,v}\varepsilon^2$, hence $|(W_r - W_c)R_v| \leq C_v\varepsilon^2$ by (38). Combining the two summands in (41) with the triangle inequality yields (39).

Step 3: from margins to scores. Both scores read the logits only through the margins $\{g_c\}_{c \neq r}$: the softmax probability is $\pi_r = (\sum_c e^{-g_c})^{-1}$ with $g_r := 0$, and the rank set $\{\pi_c >$

$\pi_r\} = \{g_c < 0\}$ behind ρ_r turns on the margin signs alone. The adversary minimizing score_r may therefore lower each margin freely, $g_c \mapsto g_c - \delta_c$ with $0 \leq \delta_c \leq L_1^{(c)}\varepsilon + C_v\varepsilon^2$ from Steps 1–2; we bound the score over this box of margins and call the largest admissible drop Ψ_r .

For TPS the box has one worst corner: π_r is nonincreasing in every decrement δ_c , so lowering all margins to their budgets and recomputing the softmax is the exact worst case, with no softmax-Lipschitz constant. APS needs one more observation, because ρ_r sums over a margin-dependent set. Write the drop as $(\rho'_r - \rho_r) + u_v(\pi'_r - \pi_r)$. Lowering margins only weakens r , so $\pi'_r \leq \pi_r$ and the second term is favorable; discarding it leaves the sound bound $\text{score}_r - \text{score}'_r \leq \rho'_r - \rho_r$. The survivor ρ_r is the probability mass of competitors ranked above r , and lowering every margin to its budget makes r simultaneously weakest against all of them, maximizing that mass. The same all-lowered corner therefore dominates APS too, and evaluating Ψ_r there is a valid upper bound, with no enumeration of rank crossings. Either score then gives (40), and the competitor-worst constants give the headline form. $\square$

The shift bound is a margin budget read in two pieces. The linear constant L_1^c is the rate at which the most exposed decision margin moves, factored as the readout gap $\|W_{y_v} - W_c\|_2$ times the equilibrium sensitivity $S_{c,v}$, the same two-amplifier product as the radius r_v but expressed in score units. The quadratic constant C_v is the price of leaving the linear regime, the curvature $L_J/(2(1-\kappa)^2)$ with its two resolvents, scaled by the readout gap. Carrying the quadratic term is what makes the certificate sound rather than merely first-order: the bare $\sigma_1(S_c)\varepsilon$ screen can be breached just past r_v , whereas $L_1^c\varepsilon + C_v\varepsilon^2$ over-states the true drop for every $\varepsilon < \varepsilon_{\text{crit}}$, which is why the empirical gate is conservative (0.92–0.98 at $\varepsilon = 0.05$) rather than tight.

Remark 4 (Where rigor stops, honestly). Two load-bearing conditions remain. (1) **Affine head.** Step 1 needs f affine in z_v^* ; a nonlinear readout would replace $W_r - W_c$ by the head Jacobian and add a Lipschitz factor. Our models use a linear head, so this holds with equality. (2) **Contractive regime.** The curvature bound and the finiteness of $\|z^*\|$ both need $\kappa < 1$ and $\varepsilon < \varepsilon_{\text{crit}}$ (A3); past $\varepsilon_{\text{crit}}$ the resolvent $1/(1-\kappa)$ is meaningless and the bound is void, the same scope as Theorem 1. On the conformal subgraph $\kappa \approx 0.68$, so $(1-\kappa)^{-2} \approx 9.8$ inflates C_v , yet the gate confirms the bound is not breached. The earlier worry about “omitted higher-order terms” does not arise: section C.1 shows the operator is bilinear on each linear region, so (22) is the complete equilibrium Hessian and C_v rests on the full second derivative, not a dominant-term reduction.

Theorem 4 (Robust coverage of AEGIS-Conformal). Let $\{(v_i, y_{v_i})\}_{i \in \text{cal}}$ be a split-conformal calibration set and v a test node. Assume:

- (C1) Exchangeability. The clean true-label scores $\{\text{score}_{y_{v_i}}(v_i)\}_{i \in \text{cal}} \cup \{\text{score}_{y_v}(v)\}$ are exchangeable (an inductive split, or a transductive split with a permutation-invariant predictor; Zargarbashi, Antonelli, and Bojchevski 2023).

(C2) Score-shift bound. For every calibration node w and the test node, simultaneously over all feasible $\|\delta\hat{A}\|_F \leq \varepsilon < \varepsilon_{\text{crit}}$, $\text{score}_{y_w}(w; \hat{A} + \delta\hat{A}) \geq \text{score}_{y_w}(w; \hat{A}) - \Delta_{y_w}(\varepsilon)$ with Δ_{y_w} the bound of Lemma 1.

Form the calibration-robust threshold

$$\hat{q}_{\text{rob}} = \text{Quantile}_{\lceil (n_{\text{cal}}+1)(1-\alpha) \rceil / n_{\text{cal}}} \left(\left\{ \text{score}_{y_{v_i}}(v_i; \hat{A}) - \Delta_{y_{v_i}}(\varepsilon) \right\}_{i \in \text{cal}} \right), \quad (42)$$

and the robust set $C_\varepsilon(v; \hat{A}) = \{r : \text{score}_r(v; \hat{A}) \geq \hat{q}_{\text{rob}}\}$. Then

$$\Pr[y_v \in C_\varepsilon(v; \hat{A} + \delta\hat{A})] \geq 1 - \alpha \quad \text{for all feasible } \|\delta\hat{A}\|_F \leq \varepsilon. \quad (43)$$

Proof. Step 1: reduce to lowered scores. Define the lowered scores $\text{score}_{y_w}(w) := \text{score}_{y_w}(w; \hat{A}) - \Delta_{y_w}(\varepsilon)$ on calibration and test. By (C2), the realized perturbed true-label score dominates its lowered clean value, $\text{score}_{y_w}(w; \hat{A} + \delta\hat{A}) \geq \text{score}_{y_w}(w)$, for every feasible $\delta\hat{A}$ at once. This is exactly the guaranteed lower envelope the binary certificate of Zargarbashi, Antonelli, and Bojchevski (2023) requires; Lemma 1 supplies it analytically for the continuous ε -ball, replacing their combinatorial enumeration of a discrete neighbourhood.

Step 2: the lowered scores stay exchangeable. Each node’s own $(W_{y_w} - W_c, S_{c,w}, L_{J,w})$, a per-point measurable transform of its clean score context, and the tie-break $u_{(\cdot)}$ is shared. Under (C1) the clean scores are exchangeable, and a common per-point transform preserves exchangeability, so the lowered scores are exchangeable too.

Step 3: split-conformal validity on the lowered scores. By the standard split-conformal guarantee (Vovk, Gammerman, and Shafer 2005) applied to the exchangeable lowered scores, with $\hat{q}_{\text{rob}}$ the $\lceil (n_{\text{cal}}+1)(1-\alpha) \rceil$ -th smallest calibration value (42), $\Pr[\text{score}_{y_v}(v) \geq \hat{q}_{\text{rob}}] \geq 1 - \alpha$.

Step 4: transfer to the whole ball. On the event $\{\text{score}_{y_v}(v) \geq \hat{q}_{\text{rob}}\}$, Step 1 gives $\text{score}_{y_v}(v; \hat{A} + \delta\hat{A}) \geq \text{score}_{y_v}(v) \geq \hat{q}_{\text{rob}}$ for every feasible $\delta\hat{A}$ simultaneously, because the envelope is uniform over the ball rather than per- δ . Hence $y_v \in C_\varepsilon(v; \hat{A} + \delta\hat{A})$ on that event, and (43) follows from the bound of Step 3. At $\varepsilon = 0$, $\Delta \equiv 0$ and the statement reduces to the ordinary split-conformal $1 - \alpha$ guarantee, as it must. $\square$

The certificate buys robustness by paying the worst-case score drop up front, on the calibration set. Lowering every calibration true-label score by its own analytic $\Delta(\varepsilon)$ before taking the quantile raises the threshold just enough that any in-ball perturbation of the test node, which can lower its true-label score by at most its own $\Delta(\varepsilon)$, still clears the bar. Because the lowering is per-node and deterministic it commutes with exchangeability, so the only probabilistic content is the ordinary split-conformal quantile event; the uniformity in Step 4 is what upgrades “robust to a fixed attack” into “robust over the whole ball.”

Remark 5 (Honest status of the coverage guarantee). (C1) is the load-bearing hypothesis, and it does not hold for free on a single fixed transductive graph: calibration and test nodes share one realized adjacency, and an IGNN’s output at v depends on the whole graph, so the scores are not i.i.d. and exchangeability must be imposed by design (inductive sampling, or a transductive-exchangeability argument as in Zargarbashi, Antonelli, and Bojchevski 2023). We therefore state (43) as sound under (C1), with the empirical coverage (clean coverage ≈ 0.90) and the worst-case gate (0.90 at $\varepsilon = 0.01$, 0.92–0.98 at $\varepsilon = 0.05$, zero equilibrium divergence across all 4138 gate nodes) as evidence that (C1) holds well enough on these graphs. The reduction to Zargarbashi, Antonelli, and Bojchevski (2023) is faithful: their construction yields robust coverage from any uniform worst-case score envelope, and Lemma 1 is exactly such an envelope, computed analytically with zero Monte-Carlo smoothing.

F Detailed Experimental Results

This section gives the detailed tables and per-dataset breakdowns behind the headline numbers of section 5, followed by the fraud-detector walk-through and the reproducibility note. All results use the 10 fixed seeds of appendix H.

F.1 Four-Quadrant Attack Comparison

Table 3 evaluates AEGIS across the four quadrants of (gradient-based, gradient-free) $\times$ (equilibrium-shift, classification-loss). The SVD direction maximizes first-order equilibrium shift by construction (Proposition 1); what matters is non-circularity and cost. A transfer direction from a separately trained surrogate (zero shared gradients) recovers 99% of the one-query damage ($\cos=0.99$), versus $44 \pm 4\%$ for a 512-query black-box search, so the fault line is model-intrinsic rather than a gradient artifact. Per query it delivers $74\text{--}156\times$ the damage of a 50-step PGD (Madry et al. 2018); classification-loss PGD inflicts 15–70% less at $50\times$ the cost, IFT-gradient PGD (solver validation, not an independent baseline) 72–92%, and prediction flips stay 0–1.8% for all methods. Figure 9 visualizes the equilibrium-damage column.

Radius vs. smoothing. Randomized smoothing (Bojchevski, Klicpera, and Günnemann 2020) gives constant per-node base radii (0.123 vs. AEGIS’s 0.078 at $\sigma=0.05$), and localized smoothing (Schuchardt et al. 2023) needs $\sim 10^5$ MC samples per node; both lack edge structure. AEGIS radii are instead structurally informative ($r_v \approx 0.10$ in dense regions, 0.01 at boundaries), a complementary point on the certificate-versus-diagnostic frontier.

Classification-gradient edges differ. Per-edge classification gradients anti-correlate with discrete ground truth on Cora ($\tau = -0.20 \pm 0.05$ vs. AEGIS’s $+0.32 \pm 0.04$) and stay near zero elsewhere; AEGIS achieves 2–6 $\times$ higher P@10, confirming that equilibrium and classification sensitivities are distinct vulnerability surfaces. Per-edge finite differences reproduce the S_c column-norm ranking ($\tau=0.999$), and iterative S_c recomputation closes ≤ 3 pp of the static-versus-greedy gap, so the static ranking is the headline. The four-quadrant equilibrium-damage table itself is promoted to the main text (table 3,

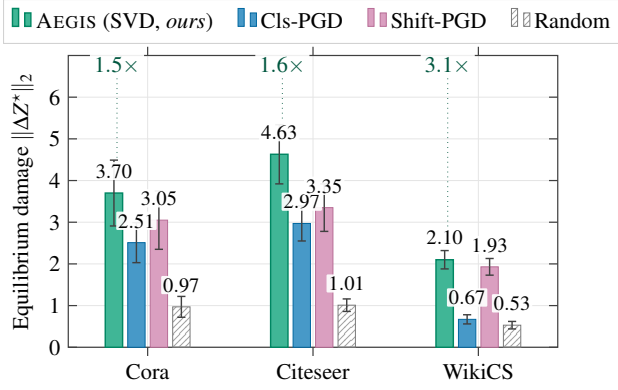


Figure 9: Equilibrium damage $\|\Delta z^*\|_2$ at $\varepsilon=0.10$ across 10 seeds, for the one-query AEGIS (SVD) attack against classification-loss PGD, IFT-gradient (shift) PGD, and a random baseline, on Cora, Citeseer, and WikiCS. Visual companion to table 3: one analytic AEGIS query inflicts 1.5–3.1 $\times$ the damage of a 50-step Cls-PGD run per dataset, at a fraction of its cost.

section 5.1); the per-edge analyses here are the supporting detail.

F.2 Comparison with Prior Structural Baselines

Table 10 and fig. 10 put AEGIS head-to-head with faithful GR-BCD and PR-BCD attackers (Geisler et al. 2021), re-implemented from the official block-coordinate algorithm and optimizing the equilibrium shift. AEGIS is a label-free one-query proxy; the question is how much of each gold-standard signal it retains. Because that ranking reads off the first-order sensitivity the iterative attackers climb by gradient, one query reaches what their 125-epoch optimizer converges to: across all six datasets and the full budget sweep $k \in \{1, \dots, 50\}$, AEGIS’s single closed-form query matches the attacker’s equilibrium damage to within $\sim 8\%$ and frequently exceeds it. It ties or beats both attackers at $k=1$ on every dataset, because the myopic greedy and relaxed attackers are not globally optimal. Rank agreement is moderate and dataset-dependent ($\tau=0.35\text{--}0.81$, mean ≈ 0.5): the orderings differ, but the damage they inflict does not. Table 11 decomposes that ranking: the sensitivity v_{ij} adds agreement over an edge-weight-only baseline on every dataset, and is decisive on three (Citeseer’s edge-weight ranking alone gives $\tau \approx 0$), so the agreement is not carried by A_{ij} . AGNNCert (Li and Wang 2025) is a sound certificate against a bounded *number* of discrete edge edits (a majority vote over hash-partitioned sub-graphs), whereas AEGIS’s r_v bounds a *continuous* edge-weight budget $\|\delta \hat{A}\|_F$; the two target different threat models and are complementary rather than comparable on one radius axis. AEGIS also beats degree-proportional, edge-betweenness, and spectral rankings at $\varepsilon=0.01$ (+6–148% AtkAdv, Wilcoxon $p < 0.001$) and inflicts 3–10 $\times$ more equilibrium damage than Mettack (Zügner and Günnemann 2019a) in the early-warning regime $k \in \{1, \dots, 5\}$ (149/150 paired wins, $p < 10^{-43}$).

Figure 1 positions these threads on a seven-axis capability radar, each polygon tracing the *frontier* of its thread (the best any cited method in it achieves on that axis). No single method covers every axis: structural attacks own attack direction,

Table 9: Threat models and outputs. Only AEGIS both names vulnerability-driving edges ($\bullet$ in “Edges”) and returns a per-node certificate (“Cert.”), from one closed-form query. The certifiers target a *discrete*-edit threat model (a bounded edit *count*), whereas AEGIS bounds a *continuous* edge-weight budget $\|\delta \hat{A}\|_F$, so they are complementary rather than comparable on a single radius axis.

Method	Threat model	Edges	Cert.
AEGIS	continuous $\ \delta \hat{A}\ _F$ ball	$\bullet$	$\bullet$
GR-BCD/PR-BCD (Geisler et al. 2021)	discrete edge edits	$\bullet$	–
AGNNCert (Li and Wang 2025)	bounded edit count	–	$\bullet$
Convex relax. (Zügner and Günnemann 2019b)	bounded perturbation	–	$\bullet$
Collective smooth. (Schuchardt et al. 2021)	random edge flips	–	$\bullet$

Table 10: AEGIS’s one query vs. faithful GR-BCD/PR-BCD (Geisler et al. 2021) (10 seeds): rank agreement τ (AEGIS’s $A_{ij}v_{ij}$ ranking vs. GR-BCD’s selection) and the damage ratio $\|\Delta z^*\|_{\text{AEGIS}} / \|\Delta z^*\|_{\text{attacker}}$ at the smallest and largest budget. Ratios near 1 are the damage-equivalence (≥ 1 means AEGIS does at least as much); per- k curves in fig. 10.

Dataset	$ E $	τ	AEGIS/GR-BCD		AEGIS/PR-BCD	
			$k=1$	$k=50$	$k=1$	$k=50$
Cora	61	+0.44	1.01	0.99	1.01	0.99
Citeseer	73	+0.37	1.00	0.97	1.00	0.96
Pubmed	53	+0.81	1.00	0.99	1.00	0.99
WikiCS	55	+0.35	1.33	1.00	1.33	0.99
Amazon	85	+0.60	1.00	0.99	1.00	0.99
Amazon Fraud	78	+0.46	1.00	1.01	1.00	0.99

per-edge ranking, and large-budget damage; the white-box gradient is the attribution oracle that AEGIS only proxies ($\tau=0.16\text{--}0.99$), while certifiers own the per-node certificate. AEGIS wins no specialist axis outright, but spans all seven, reading an attack direction, per-edge attribution, and a first-order per-node certificate off a single closed-form, label-free S_c query on an off-the-shelf model. Its third pillar, hardening via the $\sigma_1(S_c)$ penalty of section 5.3, is a different kind of claim, benchmarked there against the robust-architecture thread rather than scored on this audit-and-certify radar.

Among the certifiers it scores, AGNNCert (Li and Wang 2025) is deterministic ($T=30\text{--}80$ subgraph evaluations) and localized smoothing (Schuchardt et al. 2023) is a *collective* certificate at $\sim 10^5$ samples/node; both require specialised retraining and name no vulnerability-driving edges.

F.3 Certification vs Randomized Smoothing

The matched smoothing $\sigma=\varepsilon/\sqrt{2|E|}$ (Frobenius ball) is so small that the smoothed classifier is unchanged and abstains; only on the strictly larger per-coordinate ball ($\sigma=\varepsilon$) does it certify (0.77–0.96 of nodes), and then at **23,000–57,000 $\times$** the wall-clock of the zero-sample S_c bound. Even on the matched ball the analytic certificate is non-vacuous and **11,700–16,700 $\times$** faster: the comparison favors smoothing twice (larger ball, best matching), yet the certificate of table 5 dominates by four orders of magnitude.

F.4 Phase Transition and Scalability

Phase transition. Figure 11 sweeps $\kappa_{\max} \in [0.30, 0.99]$ on Cora (10 seeds). Even when the spectral-normalization cap is pushed to 0.99, driving the theoretical $\varepsilon_{\text{crit}}$ down 230 $\times$, the trained ReLU pattern keeps $\rho(J_z) \leq 0.42$, so the resolvent

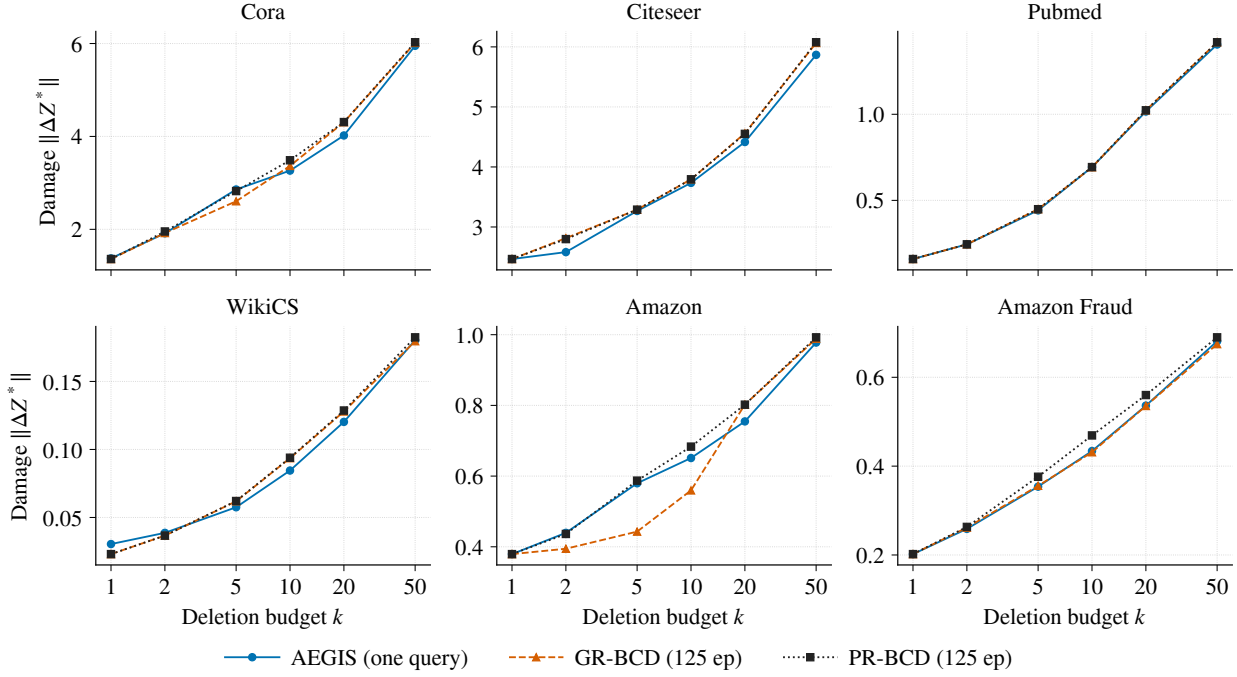


Figure 10: Damage-equivalence across the budget sweep (10 seeds). Absolute equilibrium-shift damage $\|\Delta Z^*\|$ of AEGIS’s single closed-form query against faithful 125-epoch GR-BCD and PR-BCD attackers (Geisler et al. 2021), per dataset, versus deletion budget k . AEGIS (blue) is an explicit curve; its near-coincidence with the two iterative attackers is the damage-equivalence: one label-free query recovers, and often exceeds, the equilibrium damage a 125-epoch optimizer reaches.

Table 11: Edge weight vs. sensitivity: rank agreement with GR-BCD for the full $A_{ij}v_{ij}$ ranking (τ_{Av}), edge-weight-only (τ_A), and sensitivity-only (τ_v), with the sensitivity increment $\Delta\tau = \tau_{Av} - \tau_A$ and the damage of the full ranking relative to each component (D_{Av}/D_A , D_{Av}/D_v ; means over k). The increment is positive on every dataset and the full ranking out-damages edge-weight-alone everywhere; on Citeseer/Pubmed/Amazon Fraud the sensitivity is decisive (e.g. Citeseer $\tau_A \approx 0$). These τ score agreement with the GR-BCD ranking; fig. 6 instead scores agreement with brute-force single-edge removal, where the Amazon Fraud increment reaches +0.90.

Dataset	τ_{Av}	τ_A	τ_v	$\Delta\tau$	D_{Av}/D_A	D_{Av}/D_v
Cora	+0.44	+0.29	+0.56	+0.14	1.07	1.59
Citeseer	+0.37	-0.01	+0.76	+0.39	1.15	1.26
Pubmed	+0.81	+0.63	+0.94	+0.18	1.45	1.00
WikiCS	+0.35	+0.28	+0.35	+0.07	1.03	1.11
Amazon	+0.60	+0.41	+0.59	+0.19	1.05	1.82
Amazon Fraud	+0.46	+0.19	+0.24	+0.27	1.77	1.80

grows only $1.17 \rightarrow 1.80$. This is a $2\text{--}4\times$ spectral-radius margin to criticality ($\rho=1$), the operative quantity behind $\varepsilon_{\text{crit}}$, stable across the sweep. Training converges throughout, and the matrix-free pipeline reaches its 50-step fixed-point budget at $\kappa_{\text{max}} \geq 0.85$.

Scalability and matrix-free self-consistency. Figure 12 compares the dense and matrix-free pipelines. The matrix-free $S_c v$ implements the dense S_c up to Neumann truncation; σ_1 agrees within 0.03% at $N=200$ (per-edge $\tau=0.999$), and the analytic truncation residual $\kappa^{200} \in [10^{-105}, 10^{-48}]$ across the suite (eq. (13)). For larger N the rSVD error (Halko,

Martinsson, and Tropp 2011) is bounded by the spectral gap; Pubmed ($N=19,717$) exceeds 24 GB on the dense path.

F.5 Ablations and Defense

Subgraph size. Varying $N \in \{30, 50, 100, 200\}$ on Cora (10 seeds), tightness is stable at 1.01–1.02 for $N \leq 100$ and degrades to 1.031 at $N=200$; we use $N=50$ by default ($66\times$ faster). **Full-graph scale.** A 50-node BFS covers only $\sim 1.8\%$ of a citation graph’s edges (Cora $\tau=0.16$), so at this scale we run AEGIS on the full graph, where its edge advantage over degree-proportional ranking *amplifies* to $9.82\times$ (Citeseer) and $3.25\times$ (Cora) at $k=10$, against $\approx 1.1\times$ on subgraphs. **Hyperparameters.** Tightness is stable across $d \in \{16, 32, 64, 128\}$; the spectral-norm ceiling $c \in \{0.5, 0.9\}$ traces the accuracy–robustness frontier ($r_v=0.147$ at 72.1% vs. 0.089 at 80.6%).

Edge protection and the delocalization of vulnerability. On the full graph (Cora, $N=2,708$, 10 seeds), masking the top- k edges by v_{ij} cuts $\sigma_1(S_c)$ damage by $2.4 \pm 1.8\%$ at $k=5$ and $4.6 \pm 2.9\%$ at $k=10$, a $12\text{--}46\times$ gain over random masking but far below the 42–61% a 50-node subgraph reports (where $k=5$ is already 10% of its edges). The reason is structural: the leading attack mode is *delocalized* (participation ratio 41–89 edges), so removing a handful excises a negligible slice. Reaching subgraph-scale reduction needs 2–8% of edges masked, and adaptive recomputation erodes the gain a further 1–2 pp. AEGIS therefore *localizes* risk for attribution and triage rather than yielding a few-edge defense, the failure mode that sober-look critiques (Mujkanovic et al. 2022) flag.

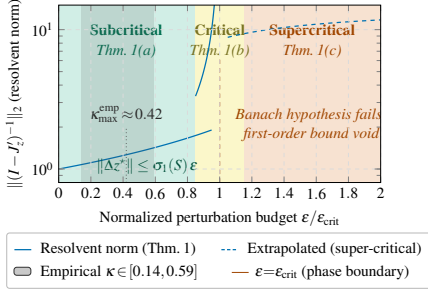


Figure 11: Three-regime characterization of Theorem 1. The empirical κ band (grey) saturates well below the spectral-normalization ceiling, so the empirical resolvent growth is far milder than the worst-case $1/(1-\kappa)$ divergence at $\varepsilon \rightarrow \varepsilon_{\text{crit}}$.

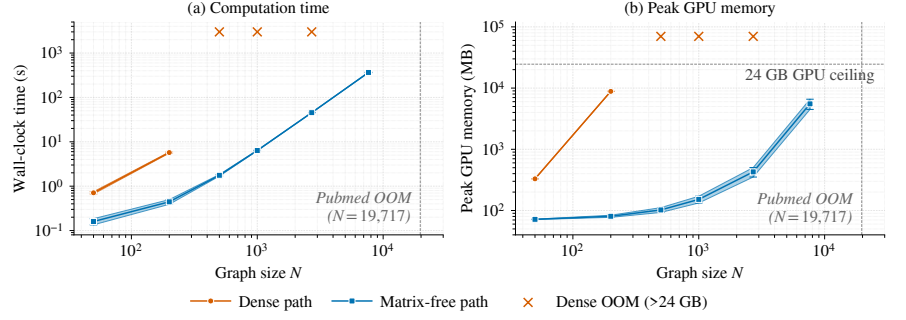


Figure 12: Dense vs. matrix-free AEGIS pipeline (RTX 4090). The dense path scales as $O((Nd)^3)$, exceeding 24 GB beyond $N=200$; the matrix-free path is roughly $O(N \log N)$ in time and sub-linear in memory, reaching Amazon Photo ($N=7,650$) at 365 s and 5.5 GB.

Unification value (one knob, three coherent effects).

Figure 13 sweeps the $\sigma_1(S_c)$ penalty λ on Cora and Citeseer (10 seeds): AEGIS’s attack damage, $\sigma_1(S_c)$ itself, and an independent GR-BCD attacker’s damage all fall together. The independent curve is the non-definitional point, since GR-BCD never sees σ_1 , yet penalizing it blunts that attacker by two orders of magnitude. Figure 14 compares the two controls on the attacker-damage versus accuracy frontier: at matched accuracy the $\sigma_1(S_c)$ penalty blunts the independent attacker more than a generic spectral cap on $\|W\|$, so the coupled operator is the more accuracy-efficient control for the attack axis. Per unit σ_1 the two are similar, since σ_1 bounds the equilibrium shift; the operator’s value is reaching a given sensitivity at higher accuracy, together with the single-knob coherence above.

The cost and completeness behind the unification are reported in table 1. The $\sim 700\times$ gap to the union of separate tools is dominated by the randomized-smoothing certificate’s 10^4 forward passes, and that union, even fully assembled, names no edge from its certificate nor any certificate from its attack, so it never delivers the per-node radius a single query reads for free.

F.6 Per-Architecture Characterization (Explicit GNNs)

For K -layer explicit GNNs (GCN, SAGE, GIN, APPNP, GAT[†] (Kipf and Welling 2017; Hamilton, Ying, and Leskovec 2017; Xu et al. 2019b; Klicpera, Bojchevski, and Günnemann 2019; Veličković et al. 2018)) we form $S_K = \partial \text{vec}(Z_K) / \partial \text{vec}(A)$ by finite differences and build S_c from S_K identically (Proposition 4). Table 12 reports the per-architecture characterization on Cora: IGNN carries Theorem 1, while the explicit models receive the computational tool without the regime characterization.

GAT[†] scope. Standard GAT uses A as a binary mask, so $\partial Z / \partial A_{ij} = 0$ on existing edges and S_c is undefined; our GAT[†] modulates attention by the edge weight ($h'_i = \sum_j \hat{A}_{ij} \alpha_{ij} W h_j$), restoring differentiability and yielding tightness 0.99, AtkAdv 4.4 $\times$, $\tau = +0.56$. Binary-mask architectures (hard attention, max/min aggregation, GATv2 (Brody,

Table 12: S_c framework on 7 GNN architectures (Cora, 10 seeds). Tight.: actual/predicted shift ratio. AtkAdv: AEGIS/random damage. τ : Kendall of the *unweighted* v_{ij} ranking vs. single-edge removal; the edge-weighted $A_{ij}v_{ij}$ ranking of Proposition 3 (fig. 6) is the higher headline score. Acc: full-graph test accuracy.

Model	Tight.	AtkAdv	τ	Acc
IGNN (eq.)	1.01 $\pm$.00	7.6 $\pm$ 0.5 $\times$	+.32 $\pm$.04	77.5 $\pm$ 1.7
GCN-2	1.00 $\pm$.00	3.6 $\pm$ 0.6 $\times$	-.04 $\pm$.03	78.9 $\pm$ 0.9
GCN-4	1.02 $\pm$.00	3.4 $\pm$ 0.4 $\times$	+.49 $\pm$.02	78.3 $\pm$ 1.8
GIN-2	1.01 $\pm$.00	2.6 $\pm$ 0.1 $\times$	+.33 $\pm$.03	76.6 $\pm$ 1.9
GAT [†]	1.01 $\pm$.01	2.1 $\pm$ 0.1 $\times$	+.54 $\pm$.06	77.8 $\pm$ 1.2
SAGE-2	1.00 $\pm$.00	1.9 $\pm$ 0.2 $\times$	+.22 $\pm$.10	77.5 $\pm$ 1.5
APPNP	1.02 $\pm$.00	3.4 $\pm$ 0.3 $\times$	+.35 $\pm$.01	82.2 $\pm$ 0.6

Alon, and Yahav 2022)-style dynamic-attention masking) fall outside the framework. **Robust backbones.** Under the $\sigma_1(W) \leq 0.9$ cap, RobustGCN-lite (Zhu et al. 2019) and GNNGuard-lite (Zhang and Zitnik 2020) match or exceed IGNN; the cap is a precondition (without it $\kappa > 1$ voids Theorem 1). **Cross-dataset transfer detail.** The edge-weighted $A_{ij}v_{ij}$ ranking beats a pure edge-weight baseline by $\Delta\tau \approx +0.16$, rising to +0.90 on the fraud graph where uniform weights are uninformative, so v_{ij} adds rank information beyond the weight. GAT-2[†] is the exception: attention reweights edges, so the unweighted v_{ij} transfers better there (+0.61 vs. +0.56). Proposition 3’s sufficient pairwise condition holds for 47–62% of edge pairs, so the global τ is empirical, not implied; where unweighted v_{ij} is uninformative (GCN-2’s near-uniform 2-hop shifts) the weighted ranking recovers the order via the edge weight (−0.28 $\rightarrow$ +0.99, Citeseer).

G Fraud-Detector Audit: Walk-Through

This section expands the audit-in-action summarized in section 6. GNN fraud detectors are a prime adversarial target: a flagged account can try to evade detection by perturbing a few of its behavioural-similarity edges (Zügner, Akbarnejad, and Günnemann 2018; Dou et al. 2020). AEGIS audits such a detector directly, with no labels and no retraining, reading from one S_c query which edges most threaten its predictions.

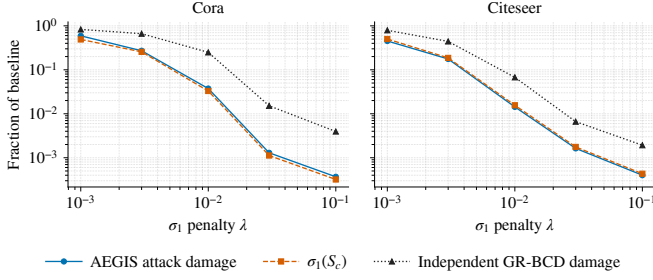


Figure 13: One knob, coherent movement: over the $\sigma_1(S_c)$ penalty sweep, AEGIS attack damage, $\sigma_1(S_c)$, and an independent GR-BCD attacker’s damage fall together (fraction of the $\lambda=0$ baseline; 10 seeds).

The fault-line diagram is fig. 7 in section 6: the trained IGNN flags the central account, and the edge-weighted ranking $A_{ij}v_{ij}$ pinpoints the fault lines, reproducing the brute-force single-edge-removal damage ranking from a single S_c query. This auditing claim is what the paper validates at scale, and non-circularly. The ranking transfers across the full fraud graph (fig. 6), precisely where uniform edge weights are uninformative so v_{ij} adds the most rank information ($\Delta\tau$ up to $+0.90$ over the weight baseline). The SVD direction transfers from a separately trained surrogate (zero shared gradients, $\cos=0.99$; section 5.1), so v_{ij} reflects model-intrinsic vulnerability rather than a gradient artifact; AEGIS out-damages discrete attackers (Mettack, GR-BCD) on their own metric in the early-warning regime (section 5.1 and appendix F.2); and the per-edge v_{ij} scores prioritize which connections to monitor, though adversarial vulnerability itself proves delocalized at scale (appendix F.5). Beyond a smoothing certifier, S_c also names *which* edges and *in which direction* (section 4.1), turning a one-query audit into an actionable defense.

H Reproducibility

All results use 10 fixed random seeds (42, 137, 271, 314, 1729, 2718, 3141, 5772, 6561, 9999); reported $\pm$ are standard deviations over these seeds. Models are trained in PyTorch (Paszke et al. 2019); IGNN (Gu et al. 2020) uses $F(Z)=\text{ReLU}(\hat{A}ZW^\top+U(X))$ with $W \in \mathbb{R}^{64 \times 64}$ spectral-normalized (Miyato et al. 2018) and Adam (learning rate 0.01). Datasets are the public splits of Cora, Citeseer, Pubmed (Sen et al. 2008), Amazon Photo (Shchur et al. 2018), WikiCS (Mernyei and Caron 2020), and Amazon Fraud (Dou et al. 2020). The randomized SVD (Halko, Martinsson, and Tropp 2011) uses $k=p=10$, $n_{\text{iter}}=2$; the Neumann series early-stops at $\|J_z^k b\| < 10^{-6} \|b\|$. Timings are on a single RTX 4090. Code release follows the disclosure protocol of section 8: the diagnostic-only path (r_v and v_{ij} scores) is released unconditionally, while the attack-direction reconstruction (algorithm 1, steps 3–4) is gated behind institutional-affiliation review.

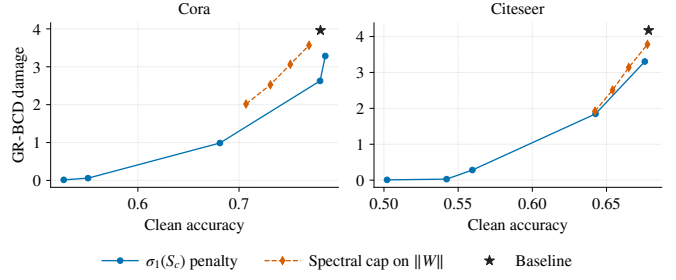


Figure 14: Independent-attacker frontier: at matched accuracy the $\sigma_1(S_c)$ penalty blunts a faithful GR-BCD attacker more than a generic spectral cap on $\|W\|$ (lower is better; 10 seeds).